

DNA Methylation

Methylation of cytosine residues in DNA plays an important role in gene regulation. DNA methylation is required for normal embryonic development. Genomic imprinting, X chromosome inactivation, chromatin modification, and silencing of endogenous retroviruses all depend on establishing and maintaining proper methylation patterns. DNA methylation is gene-specific and occurs genome-wide. Two types of methyltransferase can be distinguished by their basic functions: maintenance methylation and *de novo* methylation.

A. Maintenance methylation

This type of methylation is responsible for adding methyl groups to the newly synthesized DNA strand after replication and cell division. The methylated sites in the parental DNA (1) after replication (2) serve as template for correct methylation of the two new strands. This ensures that the previous methylation pattern is correctly maintained (3). It results in both daughter strands being methylated at the same sites as the parental DNA. The enzyme responsible for this is Dnmt1 (DNA methylase 1). Its function is essential. Mice deficient in this enzyme die as a result of genome-wide demethylation.

B. Recognition of a methylated DNA segment

Certain restriction enzymes do not cleave DNA when their recognition sequence is methylated (1). The enzyme *HpaII* cleaves DNA only when its recognition sequence 5'-CCGG-3' is not methylated (2). *MspI* recognizes the same 5'-CCGG-3' sequence irrespective of methylation and cleaves DNA at this site every time. This difference in cleavage pattern results in DNA fragments of different sizes serves to distinguish the methylation pattern of the DNA.

C. DNA methylation *de novo*

This is the second type of DNA methylation. Here methyl groups are added at new positions of both strands of DNA, not just in the hemimethylated strand as in maintenance methylation shown in A. Two genes for different methyltransferases with overlapping functions in global remethylation have been identified recently: *Dnmt3a* and *Dnmt3b*. Unmethylated

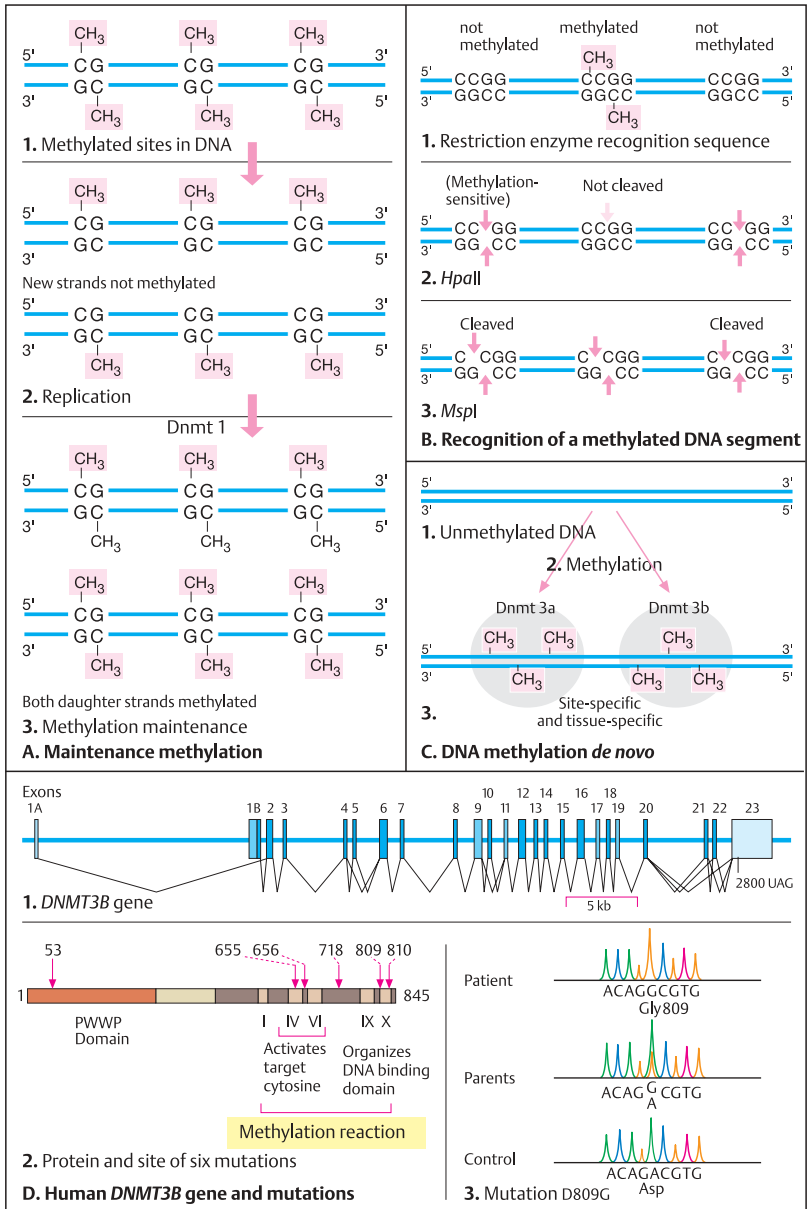
DNA (1) is methylated by their enzymes (2) in a site-specific and tissue-specific manner (3). Targeted homozygous disruption of the mouse *Dnmt3a* and *Dnmt3b* genes results in severe developmental defects. Double homozygous mutants die before day 11.5 of the 21-day embryonic development.

D. Human DNMT3B gene

Mutations in the human gene *DNMT3B* encoding type 3B *de novo* methyltransferase cause a distinctive disease called ICF syndrome (immunodeficiency, centromeric chromosomal instability, and facial anomalies, McKusick catalog no. 242860). The centromeres of chromosomes 1, 9 and 16, where satellite DNA types 2 and 3 are located, are unstable. Classical satellite DNA is grossly undermethylated in all tissues. The human gene (1) consists of 23 exons spanning 47 kb of genomic DNA. Six exons are subject to alternative splicing. The protein (2) consists of 845 amino acids with five DNA methyltransferase motifs (I, IV, V, IX, X) in the C-terminal region. The arrows point to six different mutations. The mutation at position 809 (3), a change of A to G in codon 809, i.e., GAC (Asp) to GGC (Gly), leads to the replacement of asparagine (Asn) by glycine (Gly). Both parents are heterozygous for this mutation. (Figure adapted from Xu et al., 1999).

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Genomic Imprinting

Genetic contributions from both the maternal and the paternal sets of chromosomes are necessary for a mammalian zygote to develop normally. The reason lies in a parent-of-origin-specific expression of certain genes. The genome contains defined regions where only the maternal gene copy is expressed, and not the paternal copy, and vice versa. This allele-specific gene expression, depending on the parental origin, results from so-called genomic imprinting. Genomic imprinting has important implications for human genetic disease (see p. 398).

A. The importance of two different parental genomes

Different developmental results are observed when the female pronucleus is removed from a mouse zygote (1) before the pronuclei have fused and is replaced either by another male pronucleus (2) or by a control (i.e., the same as removed) (3), or when the male pronucleus is removed and replaced by a female pronucleus (4) or a control. If the female pronucleus is replaced by a male pronucleus (2), the zygote first appears normal. However, after implantation should ensue, nearly all androgenotes fail to complete preimplantation (2). Those few that, rarely, reach postimplantation develop completely abnormally. Such embryos do not develop beyond the 12-somite stage. When a male pronucleus is replaced by a female pronucleus (4), a gynogenetic zygote, which differs markedly from the androgenote. Here, about 85% of gynogenotes reach normal preimplantation development. But although the embryo at first develops fairly well, the extra-embryonic membranes are absent or underdeveloped, and the gynogenote dies before reaching the 40-somite stage. (Figure redrawn from Sapienza and Hall, 1995).

B. Human embryonic development depends on the presence of a maternal and a paternal genome

A naturally occurring human androgenetic zygote is a hydatidiform mole (1). This is an abnormal placental formation containing two sets of paternal chromosomes and none from the mother. No embryo develops although implantation takes place. The placental tissues develop

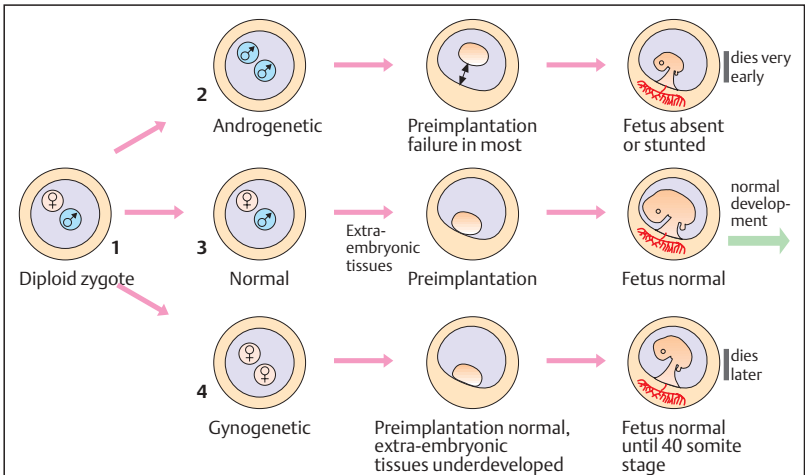
many cysts (2). When only maternal chromosomes are present, an ovarian teratoma with many different types of fetal tissues develops (3). No placental tissue is present in this naturally occurring gynogenetic zygote. In triploidy, a relatively frequent global chromosomal lethal human disorder (see p. 402), extreme hypoplasia of the placenta and fetus is observed when the additional chromosomal set is of maternal origin (4). (Photographs kindly provided by Professor Helga Rehder, Marburg.)

C. Genomic imprinting is established in early embryonic development

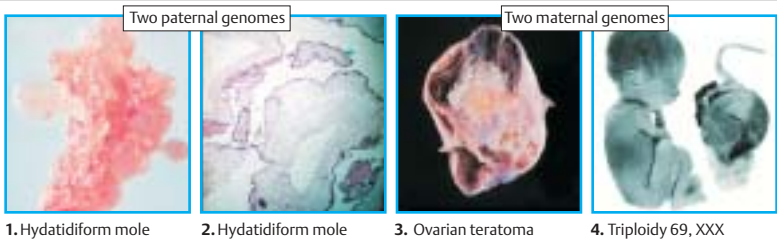
The imprint pattern present in somatic cells (1), with one allele active only either, the maternal or the paternal, propagated through all mitotic divisions. However, in primordial germ cells the imprint is erased (2). During gametogenesis the imprint is reset (3). In the male germline all gametes receive the paternal imprint and in the female germline all gametes receive the maternal imprint. After fertilization, the correct imprint pattern is present in the zygote (4). It is maintained through all subsequent cell divisions under the control of a regional imprinting center (see p. 398).

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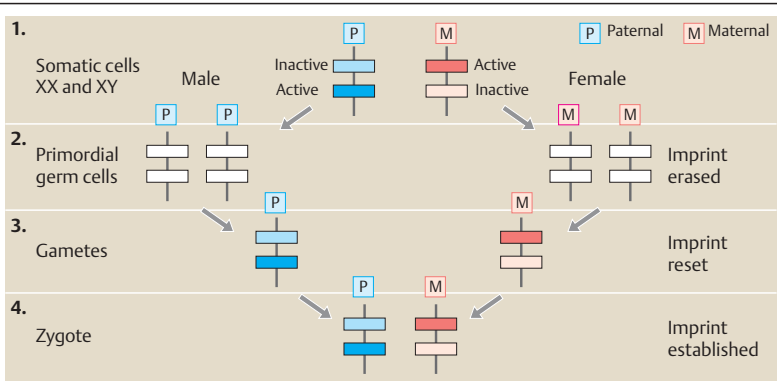
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A. The importance of two different parental genomes



B. Human embryonic development depends on presence of a maternal and a paternal genome



C. Genomic imprinting is established in early embryonic development

X-Chromosome Inactivation

During early embryonic development of mammalian females, one of the two X chromosomes becomes inactivated. The inactivation is induced by a gene (*XIST*, X-inactivation-specific transcript) on the proximal long arm of the X chromosome, transcribed from the inactive X. X inactivation is a mechanism to balance X-chromosomal gene expression between female and male cells.

A. X chromatin

In 1949, Barr und Bertram observed a stainable appendage in the nucleus of nerve cells of female (1 and 3) but not of male cats (2). The authors named this structure as “sex chromatin”. Similar structures were found drumsticks in peripheral blood leukocytes (4) and small peripheral bodies in the nuclei of fibroblasts and oral mucosa cells (5) in humans. Each of these structures represents one of the two X chromosomes and are referred to as X chromatin. (Figures 1–3 from Barr and Bertram, 1949).

B. Scheme of X inactivation

Random inactivation of most of the genes of one of the two X chromosomes in female cells occurs early in embryogenesis, at about day 21 in humans. In a given cell, it involves the X chromosome of either maternal or paternal origin. The inactivation pattern is normally irreversible and stably transmitted to all daughter cells. The expected distribution is usually 1 : 1. In rare instances this may be skewed toward a preferential type of inactivation. In extreme cases this may result in clinical manifestation in a heterozygous female if the majority of cells contain the mutation on the active X chromosome.

C. Mosaic pattern of expression

Female somatic tissues show a mosaiclike distribution of cells expressing just one of the two alleles. In mice, X-chromosomal coat-color mutants show a mosaic of light- and dark-colored coat patches (1, after Thompson, 1965). In humans, a similar distribution of normal and absent sweat pores is seen in female heterozygotes for hypohidrotic ectodermal dysplasia. The sweat pores of hemizygotes are absent (hypohidrosis). Fingerprints of female heterozygotes show areas with normal sweat pores

(black points) and areas with absent sweat pores (2a and 2b, from Passarge and Fries, 1973). In cell cultures from female heterozygotes for X-chromosomal HGPRT deficiency (hypoxanthine–guaninephosphoribosyltransferase) the colonies are either HGPRT⁻ or HGPRT⁺ (3). (From Migeon, 1971, with kind permission of the author and publisher).

D. Exceptions from X-inactivation

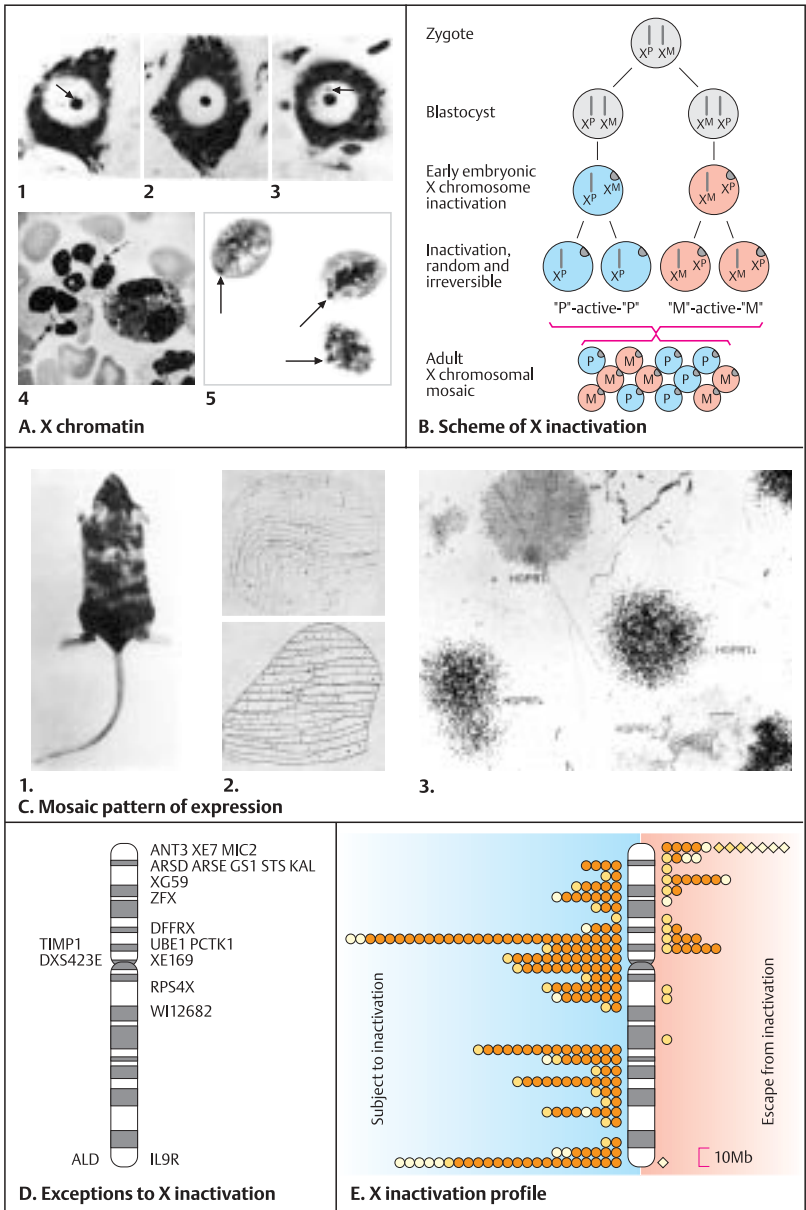
Transcriptional silencing of X-chromosomal genes in mammalian female cells is not complete. Some genes escape inactivation and are expressed from both the active and the inactive X chromosome. The majority of these genes have a homologue on the Y chromosome, reflecting a common evolutionary origin. Panel D shows genes that are expressed on the inactive human X chromosome. Most are located at the ends of the X chromosomes. (Figure adapted from Brown et al., 1997).

E. X-inactivation profile

An analysis of 224 X-linked transcripts showed that 34 escape inactivation (Carrel et al., 1999). Of these, 31 map to the short arm of the X chromosome. The expressed genes are open circles, the inactivated genes filled circles. Several genes in the pseudoautosomal region of Xp are shown as diamonds. (Figure adapted from Carrel et al., 1999).

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Targeted Gene Disruption in Transgenic Mice

The specific experimental inactivation of a gene affords an opportunity to study its normal function by comparing it with that of the inactive state. This will yield information about the role of a particular gene in development or other functions that would not be available otherwise. In the approach described here, a normal gene is replaced by a mutant allele by disrupting the normal gene (targeted gene disruption). The effects can be studied in different embryonic stages of mouse development and after birth. Subsequently the results can be compared to the effects of mutations in homologous human genes as seen in human genetic diseases. The method requires the use of mouse embryonic stem (ES) cells. ES cells are pluripotent, i.e., they are capable of giving rise to different kinds of cells but not to an entire organism. These cells can be grown in culture through many generations and yet retain the potential to be integrated into a mouse blastocyst. Here they participate in embryonic development allowing mice to produce that are homozygous for a mutation introduced into a specific gene (knockout mice).

A. Transgenic mouse derived from ES cells with a disrupted gene

Embryonic stem cells (ES) from a mouse blastocyst are isolated (1) after 3.5 days of gestation (of a total of 19.5 days) and transferred to a cell culture (2). Here they will grow on a layer of irradiated cells that are themselves unable to grow (feeder layer). Target DNA (see B) from a mouse homozygous for a marker coat color, e.g., dominant black, is added to the ES cell culture. Very few cells or perhaps just one may take up target DNA by recombination with the homologous gene in the ES cell (3). This disrupts the normal gene. These recombinant ES cells can be grown in a selective culture medium (nonrecombinant cells will not grow, see B). Recombinant ES cells containing a copy of the disrupted gene are injected into a recipient mouse blastocyst (4). These cells will be integrated into the early mouse embryo (5). The blastocyst partly containing recombinant ES cells is transferred into a pseudopregnant mouse (6). After birth (6), mice derived from normal and recom-

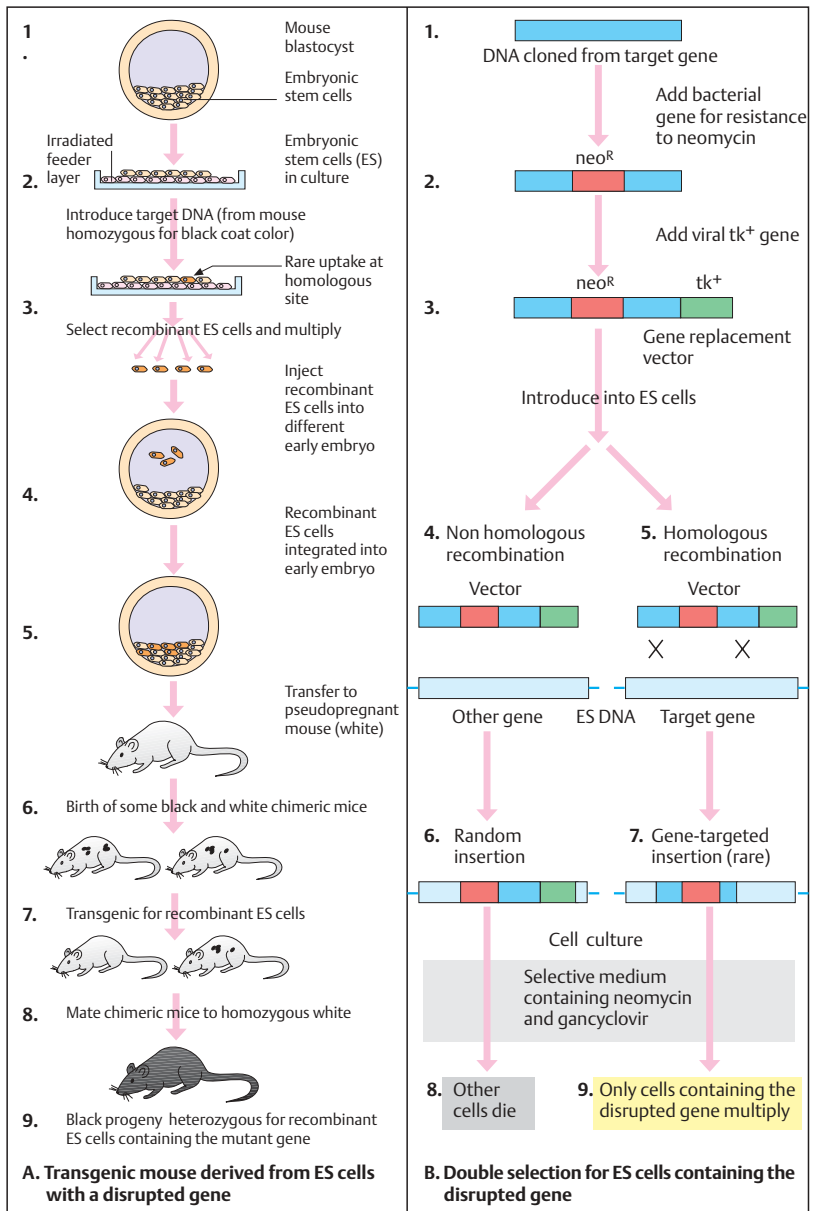
binant cells (chimeric mice) can easily be identified by spots of black coat color (7). When adult chimeric mice are mated to normal mice homozygous for another coat color allele (e.g., white, 8), the birth of black progeny indicates that the targeted gene is present in the germ line (9). The mating of such heterozygous mice (not shown) will produce mice that are homozygous for the disrupted gene (knockout mice). (Figure redrawn from Alberts et al., 1998).

B. Double selection for ES cells containing the disrupted gene

The isolation of mouse ES cells with a gene-targeted disruption requires positive and a negative selection. A bacterial gene conferring resistance to neomycin (*neo^R*) is added (2) into DNA cloned from the target gene (1). DNA containing the thymidine kinase gene (*tk⁺*) from herpes simplex virus is then also added to the vector outside the region of homology. These two selectable modifications (*neo^R* and *tk⁺*) are part of the replacement vector (3). The vector is introduced into ES cells, which are grown in culture. All ES cells that do not take up the vector (the majority) are sensitive to neomycin and will not grow in a medium containing the antibiotic (positive selection, not shown). Cells that take up the vector (about 1%) do so either by nonhomologous insertion at random sites (4) or by homologous recombination at the target site (5). These cells can be distinguished when grown in a selective medium containing gancyclovir, a nucleotide analogue toxic to cells containing *tk⁺*. Only cells containing the gene-targeted insertion (through homologous recombination) will grow because they do not contain *tk⁺* (negative selection). (Figure redrawn from Lodish et al., 2000).

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Genomics

Genomics, the Study of the Organization of Genomes

A genome contains all biological information required for life and/or reproduction.

The term genomics for the study of genomes was introduced in 1987 by V. A. McKusick and F. H. Ruddle at the suggestion of T. H. Roderick of the Jackson Laboratory, Bar Harbor, Maine, USA. The term *genomics* extends beyond *genetics*. While the latter mainly deals with heredity and its mechanism and consequences, the term genomics encompasses many aspects relating to molecular and cell biology: the different types of genomic maps; nucleic acid sequencing; assembly, storage, and management of data; gene identification; functional analysis (functional genomics), evolution of genomes; and other interdisciplinary areas relating to the wide variety of genomes in different organisms. A eukaryotic genome, contained in the chromosomes, is many times larger than a prokaryote genome. A prokaryotic genome consists of a circular chromosome with compactly arranged genes.

Important insights about the functions, evolutionary relationships, antibiotic resistance, and other metabolic aspects required for the development of new strategies for therapy are gained from sequencing the genome of microorganisms (reviewed by Fraser et al., 2000.)

A. The genome of a small bacteriophage

The genome of a bacteriophage usually consists of double-stranded DNA, although some phage genomes consist of single-stranded DNA or of RNA. The size of the genome of phages ranges from 1.6 kb to over 150 kb, representing anywhere from a few to over 200 genes. One of the smallest phages is Φ X174. F. Sanger and co-workers sequenced the genome completely and found that several of the ten genes of Φ X174 overlap. (Figure adapted from Watson et al., 1987, and Sanger et al., 1977).

B. Overlapping genes in Φ X174

The genes A and B, B and C, and D and E partially overlap. In these overlapping regions of the genome, a different reading frame is used differently by different genes. Gene E begins with the start codon ATG, the first two nu-

cleotides of which (AT) are the last two nucleotides of codon TAT for tyrosine in gene D. Similarly, the stop codon TGA for the E gene is part of codons GTG (valine) and ATG (methionine) of the D gene.

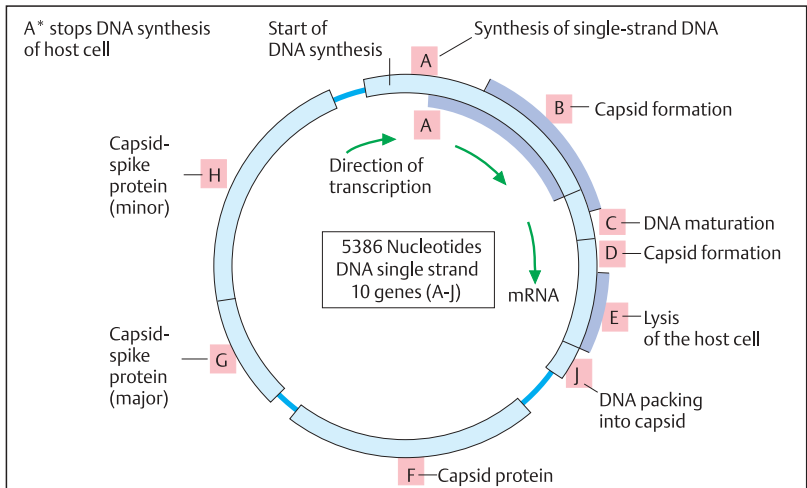
C. Genome of *Escherichia coli*

E. coli is an important microorganism. It colonizes the lower intestines of mammals including man in a symbiotic relationship. Pathogenic strains of *E. coli* cause gastrointestinal, urinary, pulmonary, and nervous system infections in humans. The *E. coli* genome has 4639221 bp. A total of 2657 protein-coding genes with known function (62% of all genes) and 1632 genes (38%) without known function have been identified. The simplified figure shows eight genes (A–H), the origin of replication (ORI), and the genes for DNA polymerase and methionine. Four operons are shown: the operons for lactose consisting of three genes, for galactose with four genes, for tryptophan with five genes, and for histidine with nine genes. About a fourth of all genes in *E. coli* are organized into 75 different operons. Most genes of the *E. coli* genome are present as a single copy; only the genes for ribosomal RNA (rRNA) are present in multiple copies. As a result, the bacteria can double their protein content every 20 minutes during cell division.

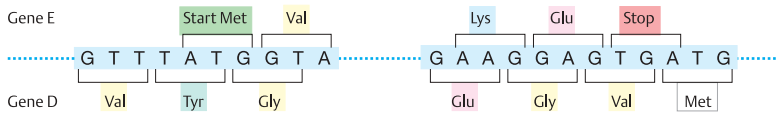
(Figure adapted from Watson et al., 1987).

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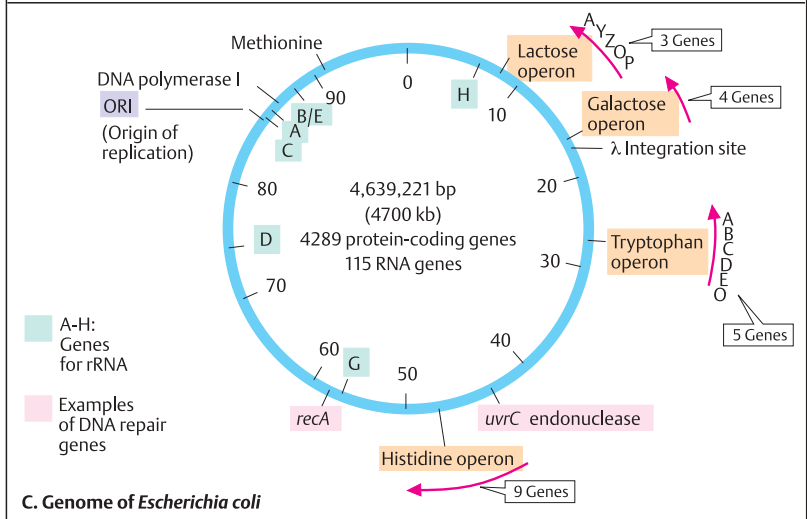
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A. Genome of the bacteriophage ϕ X174



B. Overlapping genes in ϕ X174



C. Genome of *Escherichia coli*

The Complete Sequence of the *Escherichia coli* Genome

The report of the complete sequence of the genome of the *E. coli* K-12 strain in 1997 (Blattner et al., 1997) with a full map of the 4289 protein-coding genes of the 4 639 221-base pair genome is presented here as an example of one of many sequenced microorganism genomes (see Entrez Genomes at <http://www.ncbi.nlm.nih.gov/>, and search for sequenced microorganisms). General insights into the overall structure of prokaryote genomes can be derived as outlined below. Genome sequences can be correlated with other information to identify regulatory sites, operons, mobile genetic elements, repetitive sequences, deduced functions, and homology to other organisms.

A. Overall structure and comparison with other genome sequences

The figure shows a small section of about 80 kb of the entire genome from the original publication. Base pair numbers 3 310 000 to ~3 345 000 are shown in the first row (top), and 3 339 000 to 4 025 000 in the second row. The top double line shows color-coded genes of *E. coli* encoding a protein on either of the two strands of the DNA double helix. Six other completed genomes are shown for comparison.

The average distance between genes is 118 bp. The protein-coding genes (87.8% of the genome) can be assigned to 22 functional groups (see gene function color code at the bottom of the figure). Among these are 45 genes with recognized regulatory functions (1.05% of the total); 243 genes for energy metabolism (5.67%); 115 genes for DNA replication, recombination, and repair (2.68%); 255 genes for transcription, RNA synthesis, and metabolism (5.94%); 182 genes for translation (4.24%); 131 genes for amino acid biosynthesis and metabolism (3.06%); and 58 genes for nucleotide biosynthesis and metabolism (1.35%).

Homologies between *E. coli* proteins and those of other organisms with sequenced genomes were detected in 1703 proteins of *H. influenzae* and 468 proteins of *Mycoplasma genitalium*. Yeast (*S. cerevisiae*) has 5885 proteins that match proteins in *E. coli*. 60% of *E. coli* proteins

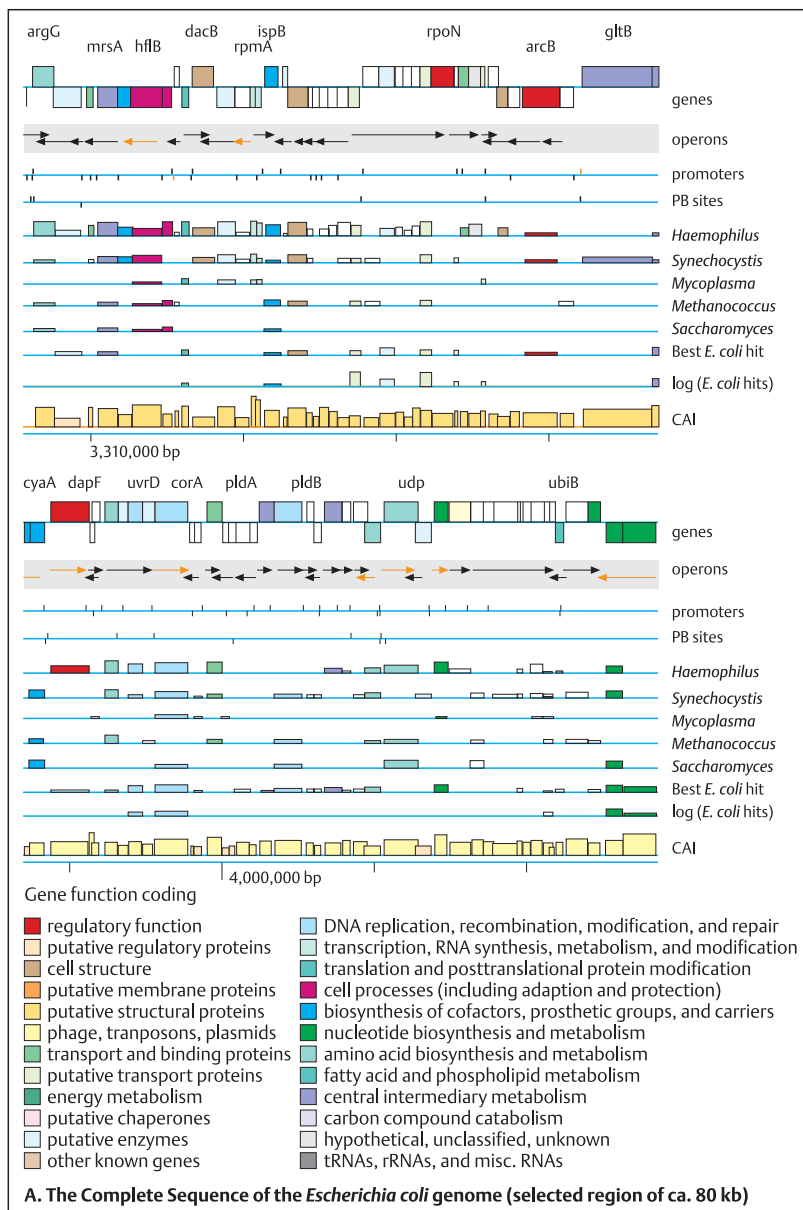
have no match in the other prokaryotes with completed genomes as of 1997.

How many genes are required for a microorganism? The smallest known cellular genome is that of *Mycoplasma genitalium*, with only 480 protein-coding genes and 37 RNA genes in 580 kb of DNA. However, not all of these genes are essential. Hutchison et al. (1999) determined by global transposon mutagenesis that 1354 of 2209 insertions into genes were not lethal to the organism. The results suggested that 265–350 of the 480 genes are essential under laboratory conditions. The limited capacity for metabolism in *M. genitalium* is compensated for by a greater dependence on the transport of molecules from the extracellular environment into the cell, mainly by ABC transporters. An ABC transporter is a heterotrimeric transport system made up of a specific ligand-binding subunit, a permease, and an ATP-binding protein.

The total of protein families required for an organism is called a *proteome*. Its study is called *proteomics*. Since many genes are required to encode the different proteins of a given metabolic or signal pathway, genes of related functions are grouped into families. Their number is considerably smaller than the total number of genes. The CAI (Codon Adaption Index) reflects the preferred codon usage of an organism. (The figure is adapted from a small part of the complete map published by Blattner et al., 1997).

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Genome of a Plasmid from a Multiresistant *Corynebacterium*

Plasmids are double-stranded circular DNA molecules in bacteria but separate from the bacterial chromosome. They are self-replicating and occur in a symbiotic or parasitic relationship with the host cell. The number of plasmids per bacterial cell varies from a few to thousands. Their sizes range from a few thousand base pairs to more than 100 kb. Plasmids usually confer a benefit to the host cell, often because they contain genes encoding enzymes that inactivate antibiotics. Drug-resistant plasmids pose a major threat to successful antibiotic therapy. Since many plasmids also contain transfer genes encoding proteins that form a macromolecular tube, or pilus, through which a copy of plasmid DNA can be transferred to other bacteria, antibiotic resistance can spread very rapidly. The following example provides new insights into the origin and evolution of a multiresistant plasmid composed of DNA segments derived from bacteria of very different origins (soil bacteria and plant, animal, and human pathogens).

A. The multiresistant plasmid pTP10

A large, 51 409 bp plasmid from the multiresistant Gram-positive *Corynebacterium striatum* M82B containing genes encoding proteins renders its host bacteria resistant to 16 antimicrobial agents from six structural classes (Tauch et al., 2000). This is the largest plasmid to have been sequenced to date. It contains DNA segments from a plasmid-encoded erythromycin (Em, shown inside the circular genome diagram) resistance region from the human pathogen *Corynebacterium diphtheriae*, a chromosomal DNA region from *Mycobacterium tuberculosis* containing tetracycline (Tc) and oxacillin resistance, a plasmid-encoded chloramphenicol (Cm) resistance region from the soil bacterium *Corynebacterium glutamicum*, and a plasmid-encoded aminoglycoside resistance to kanamycin (Km), neomycin, lividomycin, paromomycin, and ribostamycin from the fish pathogen *Pasteurella piscicida*. In addition, the plasmid contains five transposons and four insertion sequences (IS1249, IS1513, IS1250, and IS26) at eight different sites. Altogether eight genetically distinct DNA segments of different

evolutionary origin were identified.

B. Genetic map of plasmid pTP10

Plasmid pTP10 from the Gram-positive opportunistic human pathogen *Corynebacterium striatum* M82B has 47 open reading frames (ORFs). They can be assigned to eight different DNA segments forming a contiguous array of subdivided stretches in the linear representation shown (I, II, VIIb, III, VIIa, VIII, IVa, Va, VI, Vb, IVb, and VIIc).

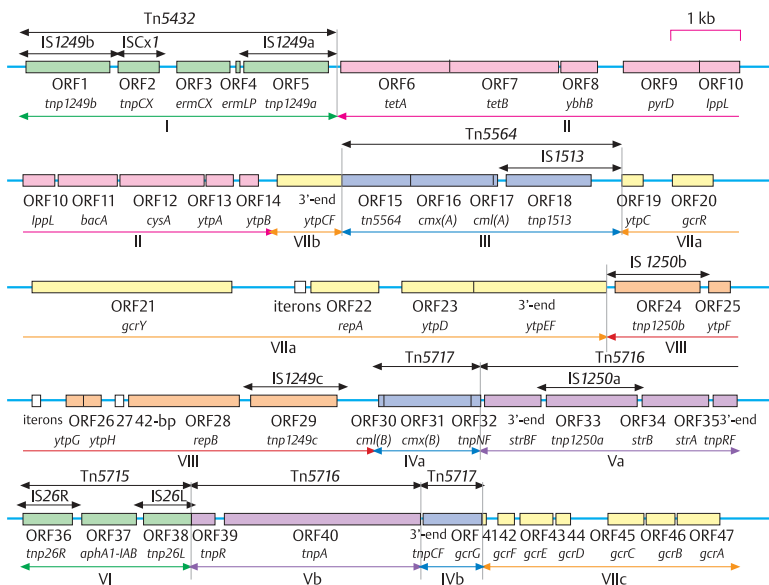
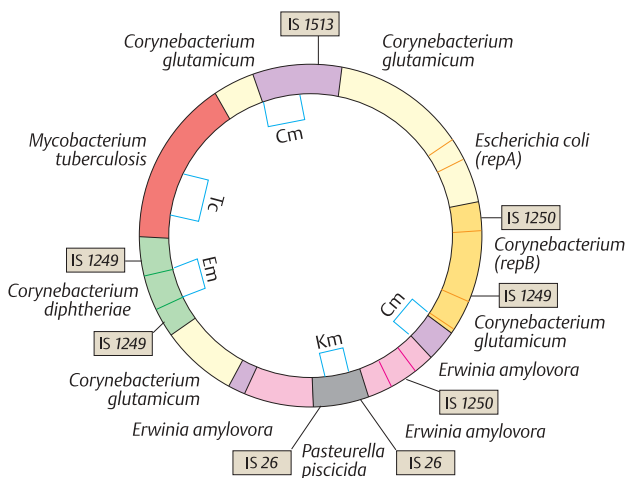
Segment I (shown in green) consists of five ORFs comprising the composite resistance transposon Tn5432. The insertion sequences IS1249b (ORF1) and IS1249a (ORF5) flank the erythromycin resistance gene region *ermCX* (ORF3) and *ermLP* (ORF4). An identical copy of IS1249 occurs in ORF29 (segment VIII). ORF3, the central region of Tn5432, encodes a 23 S rRNA methyltransferase preceded by a short leader peptide probably involved in the regulation of erythromycin-inducible translational attenuation. This region is virtually identical to the antibiotic resistance gene region (erythromycin, clindamycin) of plasmid pNG2 from *C. diphtheriae* S601.

Segment II (ORFs 6–14), located downstream of Tn5432, contains the tetracycline resistance genes *tetA* (ORF6) and *tetB* (ORF7). This segment (ORFs 6–14) is very similar to ATP-binding cassette (ABC) transporters identified in a mycobacterium (*M. smegmatis*) chromosome. The tandemly arranged genes *tetA* and *tetB* also mediate resistance to the β -lactam antibiotic oxacillin, although it is structurally and functionally unrelated to tetracycline. Presumably this results from TetAB protein heterodimerization and subsequent export of the antibiotics out of the bacterial cell. Similarly, the other segments have been delineated.

(Figures adapted from Tauch et al., 2000. I thank Professor Alfred Pühler, University of Bielefeld, for providing the figures and advice).

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Genome Maps

A genome map is a linear representation of genomic landmarks (genes and markers). It refers either to a chromosome (cytogenetic map) or to a stretch of DNA. A map provides knowledge of the position of a particular genomic landmark and its relation to others. Unraveling the human genome in some respect resembles the mapping of new continents five hundred years ago.

A *genetic map* expresses the positions of genes relative to each other without a physical anchor on the chromosome. This is the type of map that was first used by A. H. Sturtevant in 1913 when working with *Drosophila* (who started studying *Drosophila* in 1910). Here, the distance between markers is determined by the frequency of recombination during meiosis, which is in turn determined by the relative distance between the loci (see p. 116). A *physical map* provides knowledge of the exact position of a gene or marker. Its distance to another locus on the same chromosome is expressed by number of base pairs (bp), a physical equivalent. A variety of methods are employed to arrive at a physical map.

A. Physical and genetic gene maps

The physical map gives the position of a gene locus and its distance from other genes on the same chromosome in absolute values, expressed in base pairs and related to given positions along the chromosome. The genetic map gives the relative position of gene loci according to the frequency of recombination, expressed as recombination units or centimorgans (cM). One centimorgan corresponds to a recombination frequency of 1%. Since recombination occurs almost twice as often in oocytes as in spermatoocytes, the genetic map in females is about 40% longer than in males. Each gene locus has an official designation with a defined abbreviation using the letter D (for DNA), the number of the chromosome, and the number of the marker, preceded by an S for single-copy DNA, e.g., D1S77. (Diagram adapted from Watson et al., 1992).

B. STS mapping from a clone library

STS mapping plays a major role in genome mapping. An STS (sequence-tagged site) is a short stretch (60–1000 bp) of a unique DNA nu-

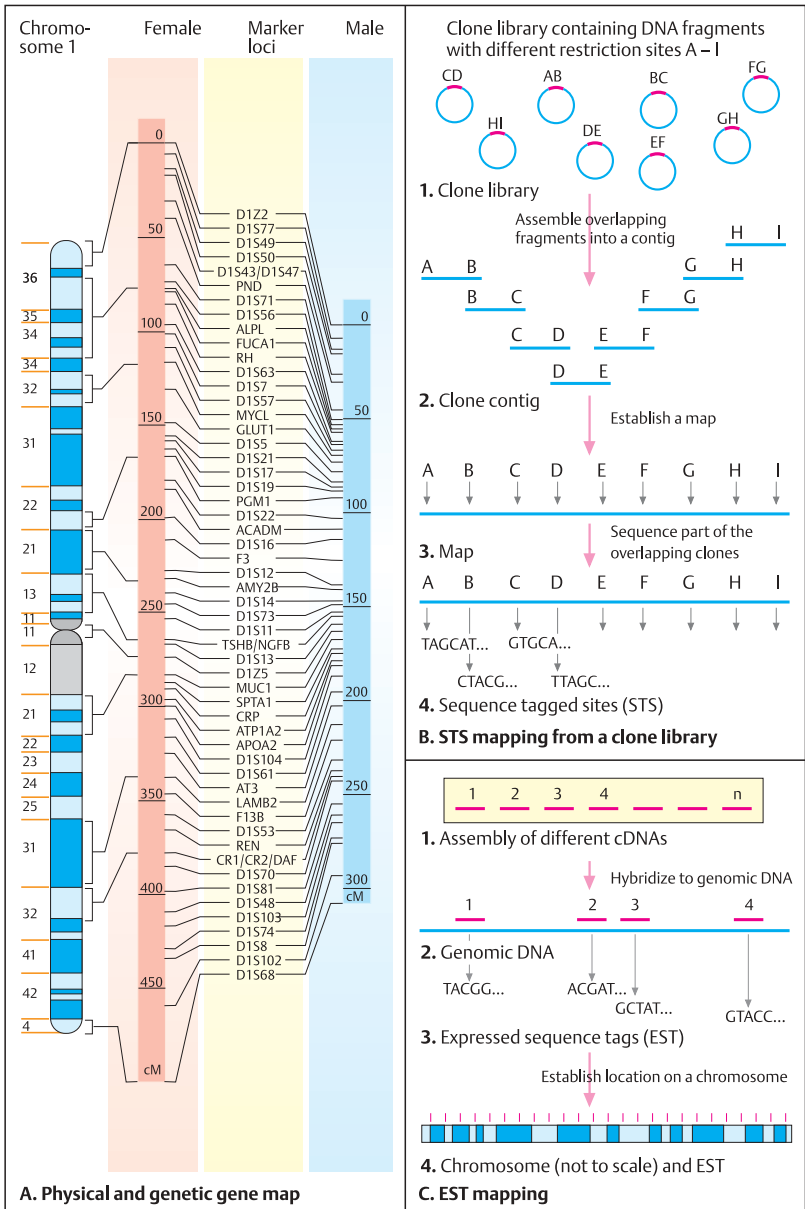
cleotide sequence. An STS has a specific location and can be analyzed by PCR (see p. 66). The relevant information, i.e., the sequence of the oligonucleotide primers used for the PCR reaction and other data can be stored electronically and does not depend on biological specimens. One can start with a clone library containing DNA fragments in unknown order (1). Each end of the chromosomal fragment is characterized by a pattern of restriction sites (see p. 64). The DNA fragments are ordered by determining which ends overlap, then assembling them as a contiguous array of overlapping fragments into a clone contig (2). These are linearly arranged. This establishes a map that shows the location and the physical distance of the landmarks, here A, B, C, etc. (3). Sequence-tagged sites (STSs) are generated from the two ends of the overlapping clones. This involves sequencing 100–300 bp of DNA (4). More than 40 000 STSs were characterized in the human genome in 1999 (see GenBank at <http://www.ncbi.nlm.nih.gov>).

C. EST mapping

ESTs (expressed sequence tags) are short DNA sequences obtained from cDNA clones (complementary DNA, see p. 58). Each EST represents part of a gene. Their location is determined by hybridizing an assembly of different cDNAs (1) to genomic DNA (2). Thus, the locations of defined sequences of expressed genes can be determined (3). These can be mapped to a location on a chromosome to establish an EST map.

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Approach to Genome Analysis

The approach to genome analysis encompasses several goals. Of primary interest is the number, type, and distribution of genes. Knowing all genes and their positions and structures in a eukaryotic genome will provide the basis for understanding their function. The size of a genome needs to be taken into account for a systematic study.

Two basic approaches to sequencing a genome can be distinguished: clone-by-clone sequencing and the so-called shotgun approach. In the former, individual DNA clones of known relation to each other are isolated, arranged in their proper alignment, and sequenced. The shotgun approach breaks the genome into millions of fragments of unknown relation. The individual DNA clones, for which prior knowledge of their precise origin is lacking, are sequenced. Subsequently, they are aligned by high-capacity computers. The two approaches complement each other.

A. Sizes of genomes and cloning vectors

The sizes of genomes of different organisms vary considerably. In general, genome size reflects the complexity of the organism. A mammalian genome (human and mouse are known best) contains 3×10^9 base pairs (bp) or 3000 Mb. If each nucleotide pair were represented by a 1-mm-wide letter, the text would be more than 3000 km long or take up more than ten sets of the Encyclopaedia Britannica or 750 megabytes of computer capacity. Thus, finding all genes, mapping their position, and determining their structure and function is an enormous task (see Human Genome Project). By comparison, the genome of important model organisms such as *Drosophila*, the nematode *C. elegans*, yeast, and bacteria are much smaller. The genomes of some important plants such as maize, rice, and wheat are even larger (5000–17000 Mb) than mammalian genomes. Since the size of DNA fragments that can be isolated and multiplied in cloning vectors for analysis is relatively small, a huge cloning capacity is necessary for analysis of a large genome. Yeast artificial chromosomes (YAC) can accommodate about 1.4 Mb, bacterial artificial chromosomes (BAC) about 0.5 Mb, whereas bacteriophages and cosmids (a phage cloning

vector consisting of the λ COS site inserted into a phage) take up only 25–50 kb of foreign DNA for cloning. Thus, vectors that clone large fragments are desirable for analyzing large genomes.

B. Range of resolution within the genome

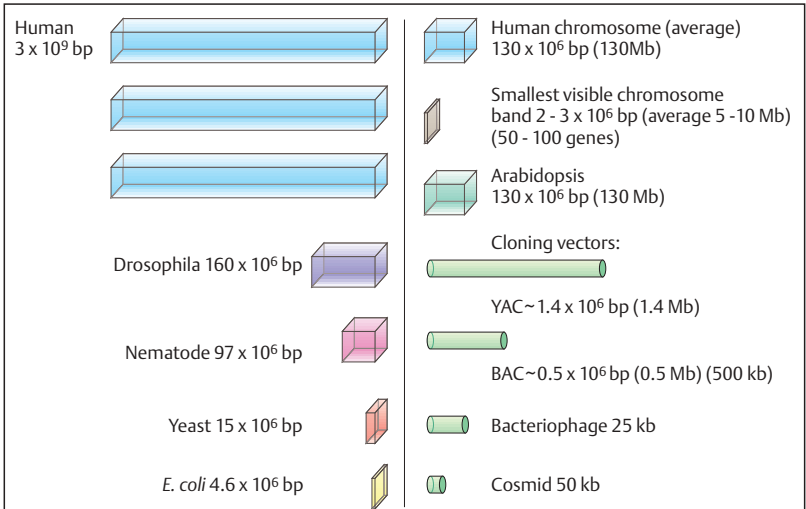
The resolution ranges from a whole chromosome or part of a chromosome isolated from a somatic hybrid cell line (1) to the sequence of the nucleotide pairs (5) and cloned DNA fragments (2). Each fragment is characterized by distinct landmarks (restriction sites or sequence-tagged sites, STS, see Genome Maps). They are aligned according to their contiguous linear orientation in a contig (3), which can be mapped (4). The individual clones can be sequenced (sequence map, 5). This approach is called a clone-by-clone approach in contrast to “shotgun sequencing” (not shown).

C. Alignment of overlapping DNA clones

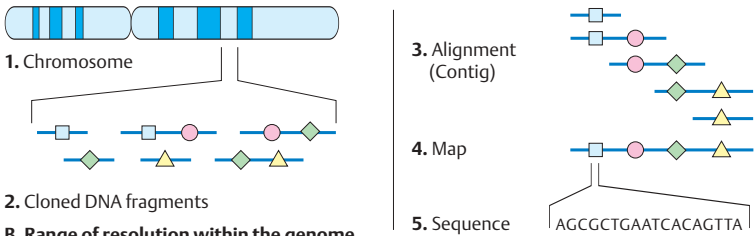
From a stretch of genomic DNA to be characterized by schematic segments A–E (1), a series of overlapping clones is derived (2). In the first step a radiolabeled clone is hybridized to genomic DNA (clone 1) from a DNA library. A probe (probe A) is used to find the adjacent clone by hybridizing to a new clone (clone 2). This establishes that clones 1 and 2 overlap. Similar steps follow. DNA farther away can be identified until one reaches the clone that contains the gene of interest. This procedure, called chromosome (DNA) walking, can start from several points and proceed in both directions. Modifications are used to speed up this process and to cover large stretches of contiguous DNA.

References

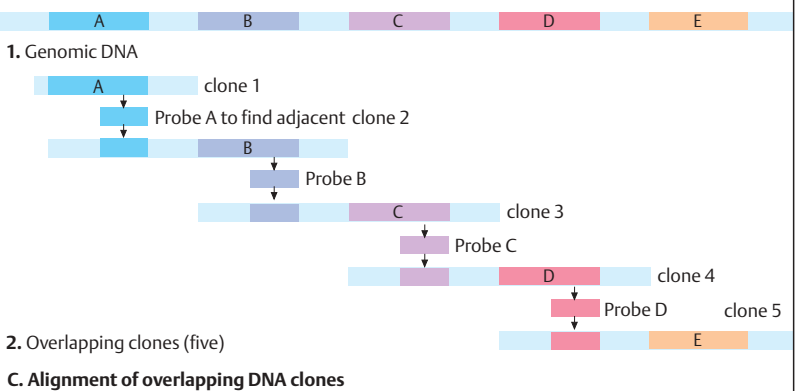
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A. Size of genomes and cloning vectors



B. Range of resolution within the genome



Organization of Eukaryotic Genomes

The genomes of higher organisms contain considerably more DNA than required for genes since most of their DNA does not have coding functions.

A. The components of the nuclear human genome

The human genome is huge, consisting of 3 billion base pairs (3×10^9 bp or 3000 Mb) per haploid set of chromosomes. Only 30% of mammalian DNA is related to genes (900 Mb), whereas 70% of the DNA is not (2100 Mb). Coding DNA in genes accounts for only 3% (90 Mb) of the total amount of DNA. The bulk of DNA (70%) consists of sequences that are repeated many times (repetitive DNA). Characteristic types of repetitive DNA are tandem repeats. Depending on their size and pattern, different types are distinguished: classic satellite DNA, minisatellites, and microsatellites. Together they constitute 14% of the total DNA (420 Mb). More than half of human DNA (56%) consists of repeats interspersed throughout the genome. The most important types are long terminal repeats (LTRs), LINEs (long interspersed nuclear elements), SINEs (short interspersed nuclear elements), and transposons. (Figure adapted from Strachan & Read, 1999).

B. Satellite DNA

When fractionated human DNA is centrifuged in a cesium chloride density gradient, the main portion of DNA (1) forms a band at buoyant density $1.701 \text{ g} \cdot \text{cm}^{-3}$. Three additional bands (satellites) appear at 1.687, 1.693, and $1.697 \text{ g} \cdot \text{cm}^{-3}$, respectively. These are less dense because their CG content differs from that of the main DNA. One distinguishes classic satellite DNA (2) made up of repeats of 100–6500 bp, minisatellites (3) of 10–20 bp repeats, and microsatellites (4) of 2–5 bp repeats. AT-rich and CG-rich segments can be recognized. Microsatellites are the most frequent form of repetitive DNA. Their general structure is $(CA)_n$ where n equals about 2–10. The human genome contains 50000–100000 polymorphic $(CA)_n$ blocks.

C. Long interspersed nuclear elements (LINEs)

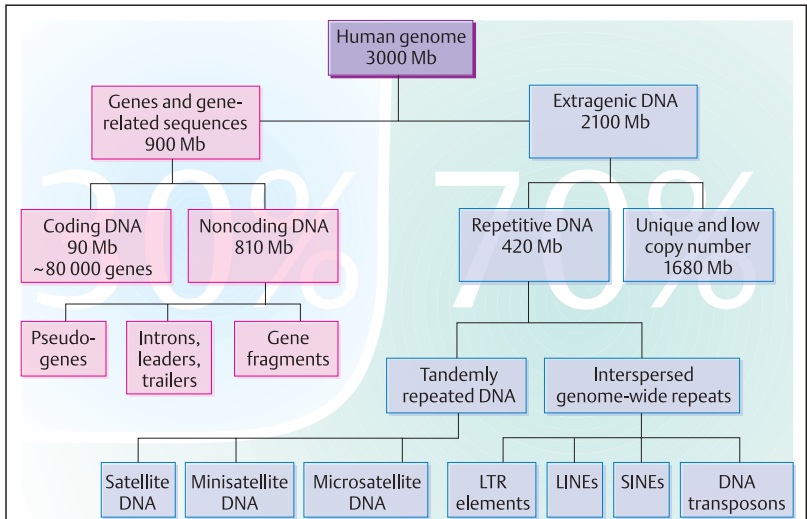
Long interspersed repeat sequences (LINEs) are mammalian retrotransposons that in contrast to retroviruses lack long terminal repeats (LTRs). They account for up to 70% of the human genome by weight. They consist of repetitive sequences up to 6500 bp long that are adenine-rich at their 3' ends. They may contain one or two open reading frames (ORF), although they are usually shorter and contain no ORF. At the 5' end and at the 3' end they have an untranslated region (5' UTR and 3' UTR). LINE elements are thought to have arisen by transposition (see p.76). Mammalian genomes contain 20000–60000 copies of LINE sequences. The major human LINE element is the L1 sequence, a segment that spans up to 6.4 kb. Approximately 100 000 L1 elements are dispersed throughout the human genome. This can result in genetic disease if one is inserted into a gene (e.g. hemophilia, p. 366).

D. Short interspersed nuclear elements (SINEs)

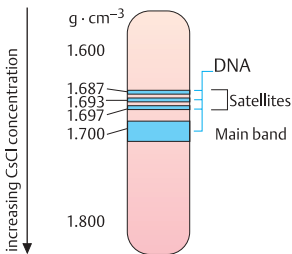
Short interspersed repeat sequences (SINEs) consist of midsize repetitive segments of similar nucleotide sequences with an average of 300 bp. Their basic structure is a tandem duplication of CG-rich segments separated by adenine-rich segments. The most frequent SINE sequences in humans are the Alu family (Alu sequences). With about 500 000 copies, they make up about 3–6% of the total genome of humans. An Alu sequence consists of two 130-bp tandem duplications with A-rich sections between them. The 3' side ("right side") contains an insertion of 32 bp.

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A. The components of the nuclear human genome



1. Cesium chloride density gradient



2. Classic satellite DNA (100-6500 bp repeats)



3. Minisatellite DNA (20-100 bp repeats)



4. Microsatellite DNA (CA)_n repeats (n = 2-10 bp)

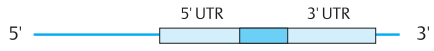
B. Satellite DNA



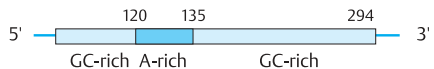
1. Full length LINE 1 (6.1 kb)



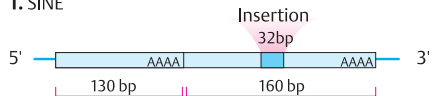
2. Truncated LINEs



C. Long interspersed nuclear elements (LINEs)



1. SINE



2. Alu dimer

D. Short interspersed nuclear elements (SINEs)

Gene Identification

A frequent and important goal in genome research is to identify a particular gene with its structure and expression and relate these to normal and abnormal function. This process has been aided by the ever-increasing number of genes with a known map location, by access to extensive data banks that store sequence information, by the availability of dense maps of marker loci, by comparative data from different organisms, and by information from other sources. The principles applied in identifying a gene are outlined here.

A. Different approaches to identifying a disease-related gene

Disease genes have been identified and isolated by different approaches, the choice of which has depended on prior information and technical considerations. The three approaches that have been useful to date are (i) positional cloning, (ii) functional cloning, and (iii) cloning of a candidate gene.

Positional cloning has been used to identify more than 20 human disease genes. The first and crucial step is clinical identification of the phenotype. Next, the putative gene is mapped to a chromosomal position with defined limits (see B). From here the gene can be isolated (cloned). In the next step, mutations in the gene are demonstrated in patients and are shown not to be present in unaffected family members and normal controls.

Functional cloning requires prior knowledge of the function of the gene. As this information is rarely available at the outset, this approach is very limited.

The candidate gene approach utilizes independent paths of information. If a gene with a function relevant to the disorder is known and has been mapped, mutations of this gene can be sought in patients. If mutations are present in the candidate gene of patients, this gene is likely to be causally related to the disease.

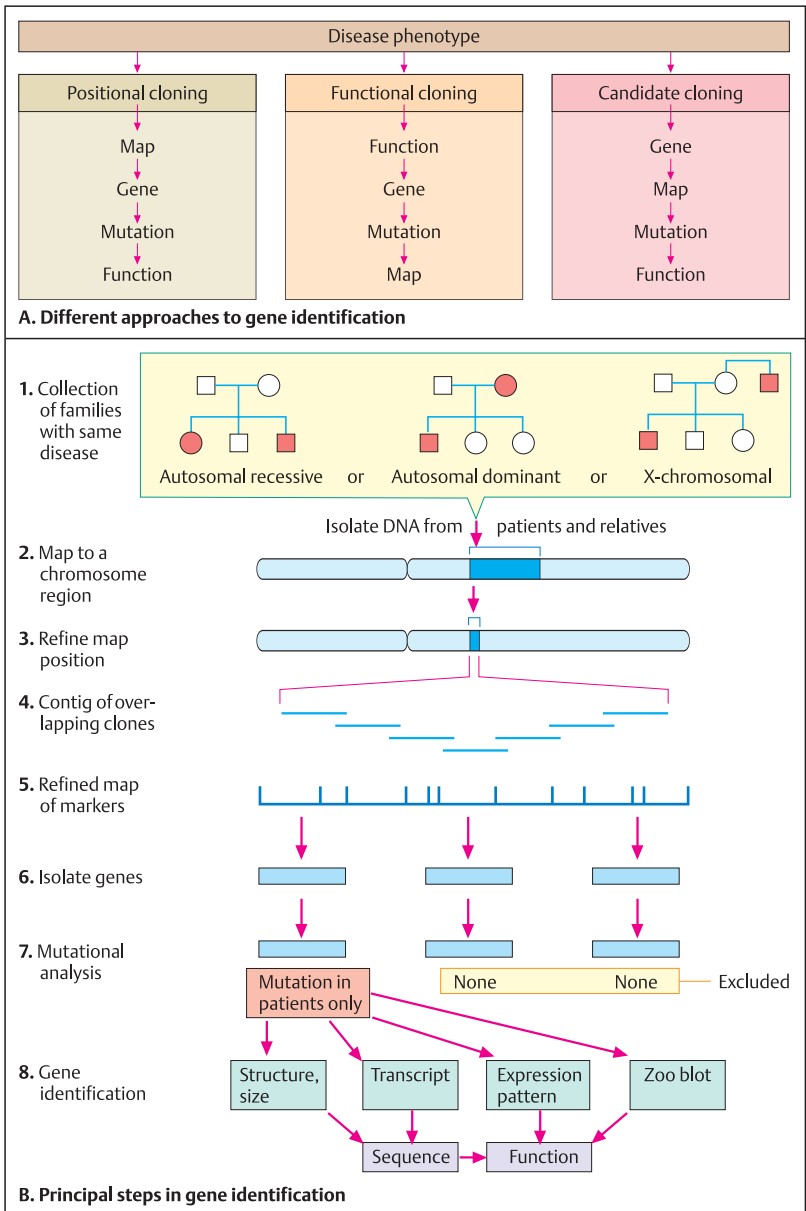
B. Principal steps in gene identification

To identify a suspected human disease gene, clinical and family data together with blood samples for DNA must be collected from patients with the same particular disorder. The disorder may follow one of the three modes of monogenic inheritance (autosomal recessive,

autosomal dominant, X-chromosomal; 1) or multigenic complex inheritance (not shown). Genetic heterogeneity must be ruled out. A chromosome region likely to harbor the gene is then identified by one of the genetic mapping techniques (linkage analysis or physical mapping using a chromosomal structural aberration such as a deletion or a translocation) (2). The map position is refined and the gene location is narrowed to a small area, e.g., a visible chromosome band within 2–3 Mb (3). A contig of overlapping DNA clones from a YAC or BAC (bacterial artificial chromosome) or a cosmid library is established from the region (4). This results in a refined molecular map (5) characterized by a set of localized polymorphic DNA marker loci (restriction map, STS map, EST map, see p. 242). The presence of open reading frames (ORF), transcripts, exons, and polyadenylated sites in this region indicates that one is dealing with a gene. Such stretches of DNA most likely representing genes are isolated and subjected to mutational analysis (6). Those genes not containing mutations in patients can be excluded (7). When mutations are found and a polymorphism is excluded, the correct gene has been identified (8). For confirmation, the expression pattern is analyzed, its structure (exons, introns) and size are determined, the transcript is analyzed, and these properties are compared with those of similar genes in other organisms ("zoo blot", see p. 250). Finally, the DNA sequence of the entire gene can be determined and conclusions about the function drawn. From here on, a gene-directed diagnostic procedure can be developed.

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The Human Genome Project

The Human Genome Project (HGP) is an international cooperative effort to investigate the human genome in its entirety. It consists of different approaches, each directed toward a different goal. Many countries participate. Five major centers, four in the United States and one in the United Kingdom, contribute about 85% of the data. The United States National Human Genome Research Institute (NHGRI: <http://www.nhgri.nih.gov/>) functions as a central agency. In addition, a number of private companies participate. The Human Genome Organization (HUGO), an international organization, is involved in many aspects of the human genome project. The principal goals of the HGP include mapping of genes and markers, sequencing the genome with special attention to genes, comparing the human genome to that of other organisms, providing new techniques, and handling of the vast amount of data. The speed of progress precludes an up-to-date presentation here. Therefore, the opposite page contains information that will give access to various aspects of the human genome project via the Internet.

A. The Human Genome Project (HGP) and related information online

Information about the Human Genome Project is best gleaned from the internet. Here, several websites provide access to detailed information much beyond the limited space available here. The opposite page (p. 249) provides information about the main areas, the HGP itself, genes and disease, gene maps, networks of databases, education, and the genomes of other organisms than man. For microbial genomes see page 234–236.

Ethical, legal, and social implications (ELSI) of the Human Genome Project

From the outset, the ethical, legal, and social implications of the human genome project have been an important consideration. About 3% of the total funding is directed to the ELSI Research program. ELSI covers a wide range of issues. These include confidentiality and fairness in the use of individual genetic information, prevention of genetic discrimination, use of genetic methods in clinical diagnostics, public and professional education, and other related issues.

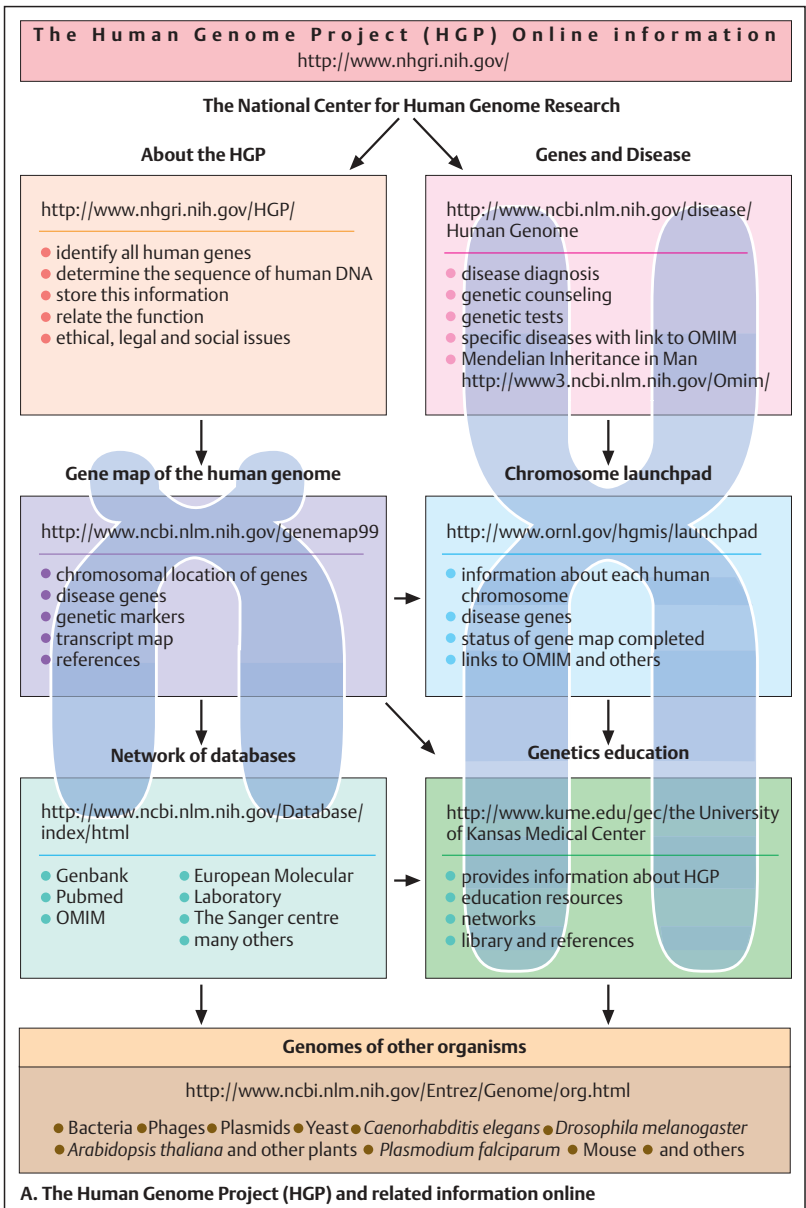
Medical implications

The human genome project carries important implications for the theory and practice of medicine. Complete knowledge of human genes will lead to better understanding of disease processes. This, in turn, will lead to improved precision of diagnosis, correct assessment of genetic risk, and the development of therapy.

Important information about disease-causing human genes is derived from studying the genomes of model organisms such as yeast, *C. elegans*, *Drosophila*, and the mouse. The recently published sequence of the *Drosophila* genome (Adams et al., 2000) revealed that of a set of 289 human genes involved in causing diseases, 177 (62%) appeared to have an orthologue in *Drosophila* (Rubin et al., 2000).

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Identification of a Coding DNA Segment

Numerous methods are available to identify a gene of interest that do not require large segments of DNA to be sequenced. Some examples are presented here.

A. Microdissection of metaphase chromosomes

If the chromosomal location of a gene of interest is known, this region can be cut out of a metaphase chromosome by means of microdissection (arrow). As first applied by B. Horsthemke and co-workers (Lüdecke et al., 1989) this method has the advantage that all other chromosomal segments are eliminated. (Photo from Buiting et al., 1990).

B. Artificial yeast chromosomes (YACs)

Large DNA fragments (200–300 kb) can be replicated in yeast cells. They are inserted into artificial yeast chromosomes (YAC, see p. 110) and replicated with them. A photograph of a transverse alternating field electrophoresis (TAFE) with nine lanes after ethidium bromide staining shows fragments of different sizes. These correspond to the naturally occurring yeast chromosomes. Six of the lanes (2–7) contain an additional band, which corresponds to an artificial yeast chromosome. They are marked with a yellow point: the lowest band in lane 2 (YAC9), the third band from the bottom in lanes 3, 4, and 6 (YAC41, YAC45, YAC51), and the lowest band in lane 7 (YAC52). In lane 5, YAC50 is masked by a yeast chromosome (third fragment from below). Lane 1 contains the standards for fragment size (preparation by K. Buiting and B. Horsthemke, Essen).

C. Exon trapping

In an unidentified segment of DNA, a gene can be recognized by the occurrence of coding segments (exons). To find and isolate an exon, the genomic fragment can be cloned in a vector that consists of a strong promoter gene and a reporter gene. An exon that is present is cut out by means of the donor and acceptor splice signals and expressed together with the genomic fragment (exon trapping). cDNA is produced from the mRNA and replicated by means of PCR. Finally, the trapped exon can be sequenced or

characterized by other means. (Diagram after Davies and Read, 1992.)

D. Single-strand conformation polymorphism (SSCP)

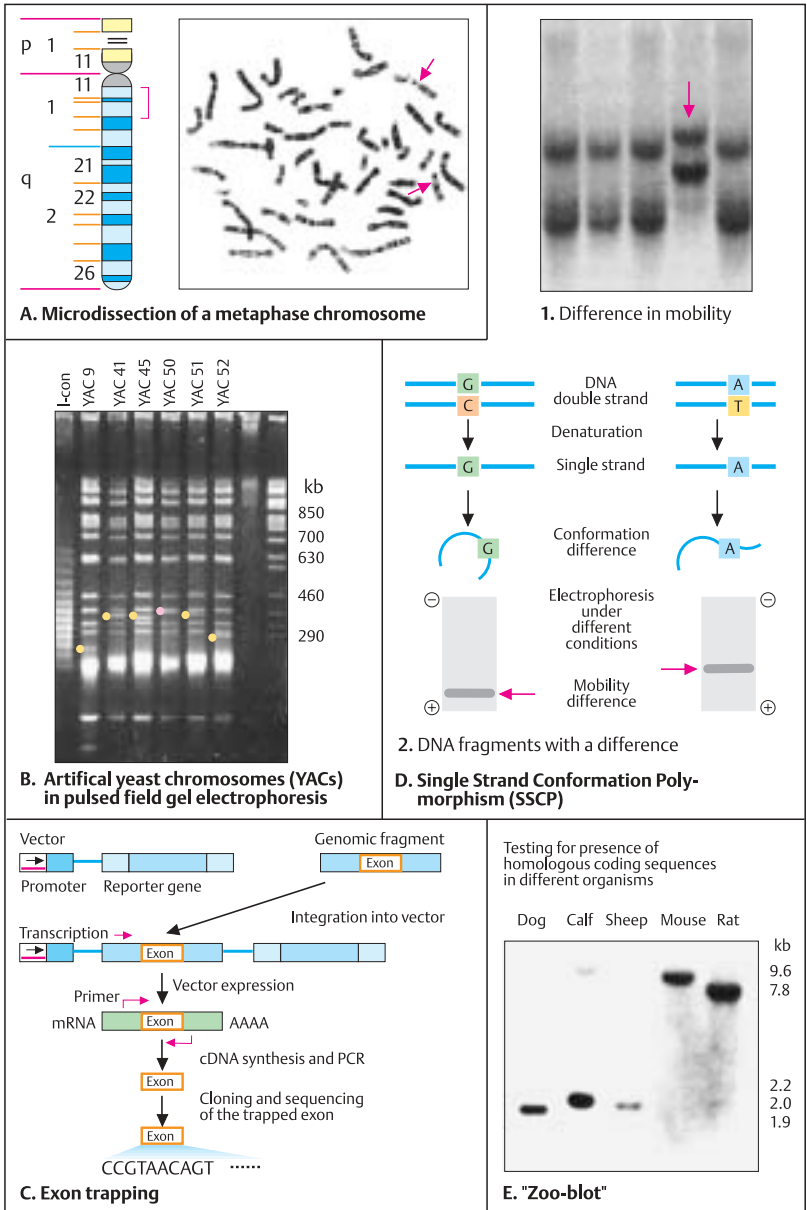
This procedure helps establish a difference in the nucleotide base sequence due to mutation or polymorphism. Whether DNA segments (single-stranded DNA) of common origin differ from each other is determined by their speed of migration in a polyacrylamide gel electrophoresis under different conditions, such as changes in temperature, in pH, etc. A base substitution may lead to a difference in spatial arrangement (conformation) and in mobility (lane 4, arrow). (Polyacrylamide gel electrophoresis and silver staining, photograph kindly provided by D. Lohmann, Essen).

E. "Zoo blot"

The cross hybridization of DNA across species boundaries ("zoo blot") is an indication for coding sequences, since genes have similar structures in different organisms. A zoo blot is a Southern blot of genomic DNA from different species. Owing to evolutionary relationship, coding DNA sequences will hybridize to the same probe. This can be taken as evidence that the DNA probed is part of a functioning gene. (Photograph of K. Buiting, Essen).

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The Dynamic Genome: Mobile Genetic Elements

In the 1950s Barbara McClintock observed an unusual phenomenon during her genetic investigations of Indian corn (maize, *Zea mays*), namely, “jumping genes” (mobile genetic elements). A mobile element causes a break in a chromosome at the site of an insertion and causes a gene locus to move to a different location in the chromosome (transposition). During the last 25 years, mobile elements have been found in every organism in which they have been sought: in bacteria, in *Drosophila*, in the nematode *C. elegans*, and in mammals, including man. These observations have resulted in the concept of a dynamic genome that is by no means fixed and unchangeable. Mutations due to mobile elements inserted into genes have also been demonstrated in man.

A. Stable and unstable mutations in Indian corn

McClintock observed not only stable mutations (e.g., violet corn kernels) but also fine or somewhat coarser pigment spots (variegation) in some kernels due to unstable mutations.

B. Effect of mutation and transposition

A gene at the C locus produces a violet pigment of the aleurone in cells of Indian corn. When a mobile element (*Ds*) inactivates this gene locus, the corn is colorless. If *Ds* is removed by transposition, C-locus function is restored and small pigmented spots appear.

C. Insertion and removal (*Ds*)

As defined by McClintock, an activator (*Ac* locus) is an element that can activate another locus, dissociation (*Ds*), and cause a break in the chromosome (1). While *Ac* can move independently (autonomous transposition), *Ds* can only move to another location of the chromosome under the influence of *Ac* (nonautonomous transposition). The C locus is inactivated by the insertion of *Ds* (2). Under the influence of *Ac*, *Ds* is then removed from some of the cells, and the C locus is returned to normal function. Since the cells of corn are of clonal origin, the time of transposition influences the phenotype. If transposition occurs early in development, the pigmented spots are relatively large; if it occurs late, the spots are small.

D. Transposons in bacteria

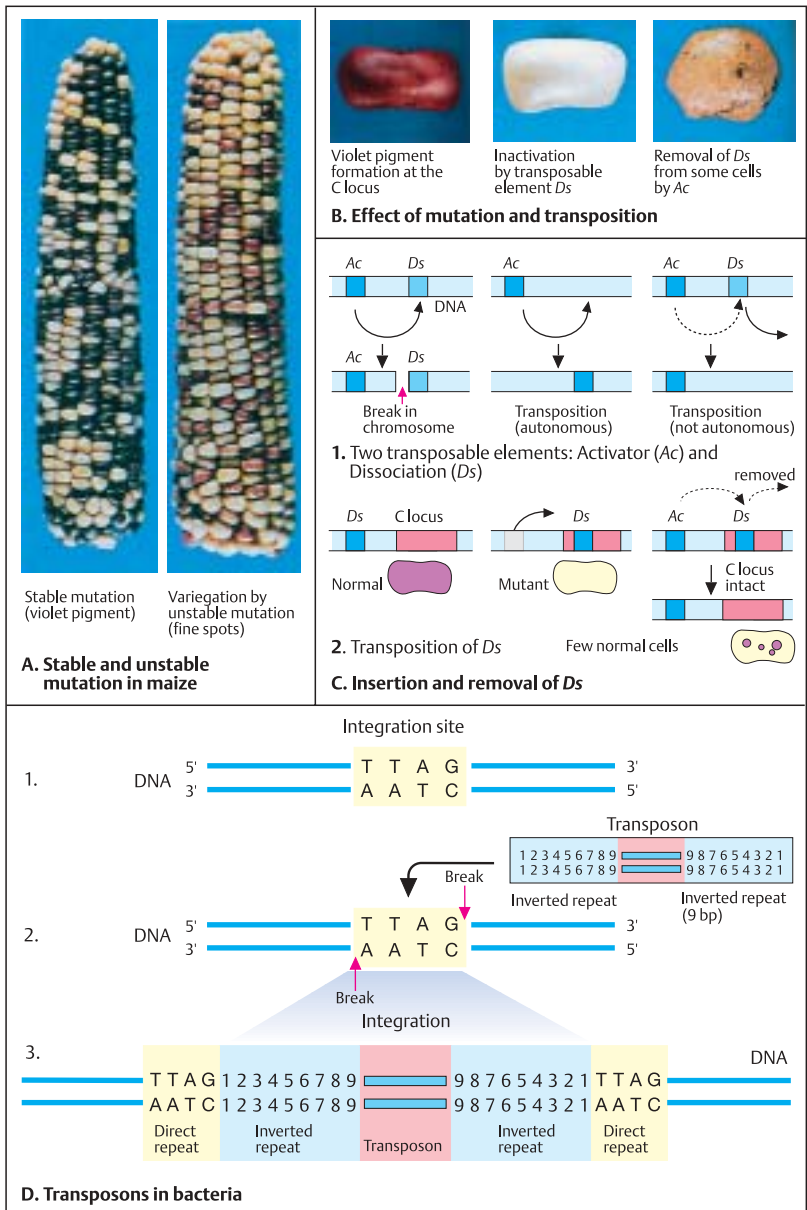
Mobile genetic elements are classified according to their effect and molecular structure: simple insertion sequences (IS) and the more complex transposons (Tn). A transposon contains additional genes, e.g., for antibiotic resistance in bacteria.

Transposition is a special type of recombination by which a DNA segment of about 750 bp to 10 kb is able to move from one position to another, either on the same or on another DNA molecule. The insertion occurs at an integration site (1) and requires a break (2) with subsequent integration (3). The sequences on either side of the integrated segment at the integration site are direct repeats. At both ends, each IS element or transposon carries inverted repeats whose lengths and base sequences are characteristic for different IS and Tn elements. The expression “direct” signifies that two copies of a sequence are oriented in the same direction (e.g., TTAG on each side of the integrated transposon). Direct and inverted repeats are evidence of the presence of a mobile genetic element. One *E. coli* cell contains on average about ten copies of such sequences. They have also been demonstrated in yeast, *drosophila*, and other eukaryotic cells.

(Photographs from N.V. Fedoroff, 1984).

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Evolution of Genes and Genomes

Genes and genomes existing today are the cumulative result of events that have taken place in the past. The classical theory of evolution as formulated by Charles Darwin in 1859 states that (i) all living organisms today have descended from organisms living in the past; (ii) the organisms living during earlier times differed from those living today; (iii) the changes were more or less gradual, with only small changes at a time; (iv) the changes usually led to divergent organisms, with the number of ancestral types of organisms being much smaller than the number of types today; and (v) all changes result from causes that continue to exist today and can thus be studied today.

A. Gene evolution by duplication

Observations on different genomes indicate that different types of duplication must have occurred: of individual genes, of parts of genes (exons), subgenomic duplications, and rare duplications of the whole genome. Duplication of a gene relieves the selective pressure on this gene. After a duplication event, the gene can accumulate mutations without compromising the original function, provided the duplicated gene attains a separate regulatory control. (Figure adapted from Strachan and Read, 1999).

B. Gene evolution by exon shuffling

The exon/intron structure of eukaryotic genes lends great evolutionary versatility. New genes can be created by placing parts of existing genes into a new context, using functional properties in a new combination. (Figure adapted from Strachan and Read, 1999).

C. Evolution of chromosomes

Evolution occurs also by structural rearrangements of the genome at the chromosomal level. Related species, e.g., mammals, differ by the number of their chromosomes and chromosomal morphology, but not by the number of genes, which often are conserved to a remarkable degree. The human chromosomes 2 and 5 are shown in comparison with the corresponding chromosomes in three closely related primates, chimpanzee, gorilla, and orangutan. The human chromosome 2 appears to have evolved

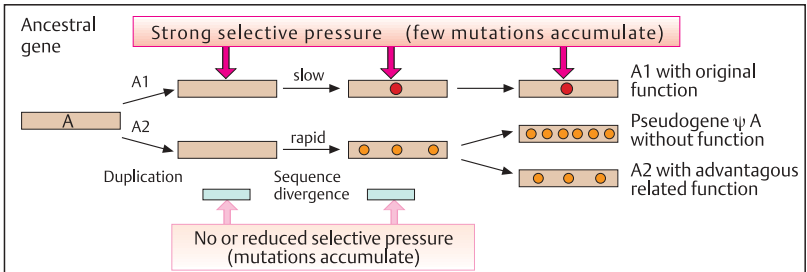
from fusion of the two primate chromosomes. The differences in chromosome 5 are much more subtle. The orangutan chromosome 5 differs from that of man and the other primates by a pericentric inversion. The banding pattern of all primate chromosomes is remarkably similar. This reflects their close evolutionary relationship. (Figure adapted from Yunis and Prakash, 1982).

D. Molecular phylogenetics and evolutionary tree reconstruction

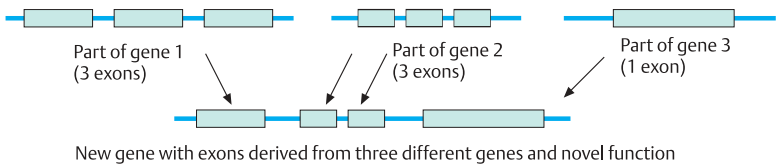
A convincing evolutionary relationship can be established by reconstructing past events. A phylogenetic tree may be based on different types of evidence: on fossils, on differences in proteins, on immunological data, on DNA–DNA hybridization, and on DNA sequence similarity. One determines the number of events that must have taken place to explain the diversity observed today. In the path from an ancestral gene (1), different events of divergence can be distinguished (two events are shown schematically here). However, trees are not always as simple as shown here. Since the term homology does not distinguish whether the common evolutionary origin is within or between species, the terms *paralogy* and *orthology* are used (2). Genes or nonallelic chromosomal segments or DNA sequences are called *paralogous* if they have evolved *within* a species (for example the two α -globin loci in humans). If they evolved *between* species prior to diversion, they are called *orthologous* (for example the α -globin and the β -globin genes).

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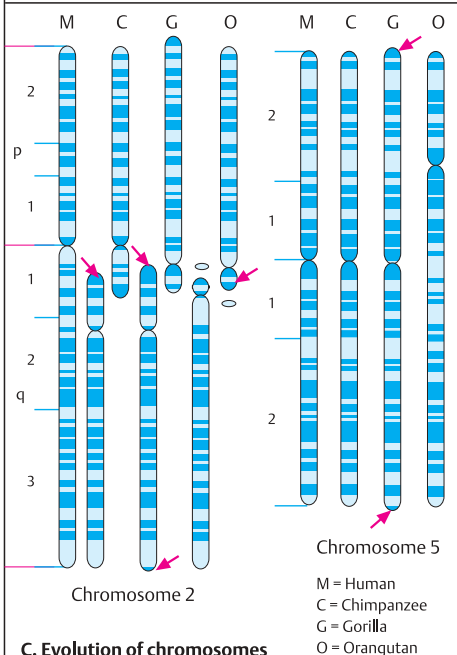
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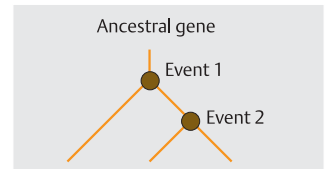
A. Gene evolution by duplication



B. Gene evolution by exon shuffling

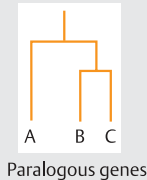


C. Evolution of chromosomes



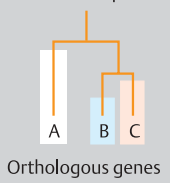
1. Three genes related by evolution (gene tree)

Within a species



2. Homology of genes by evolution between different but related species

Between different, but related species



D. Molecular phylogenetics and evolutionary tree reconstruction

Comparative Genomics

In comparing the genomes of different organisms it is interesting to determine the minimal set of gene families required to encode all proteins required for the overall function of the organism, the so-called core proteome. This is one of the goals of comparative genomics. An example for four different organisms is given in the following table.

A. Comparison of gene loci in man, mouse, cat, and cow

The conservation of certain groups of genes during mammalian evolution can be demonstrated by comparing reference loci in four different orders of mammals. Many of the loci of humans (representing the order Primates) are found in the same order, although often on different chromosomes, in the house mouse (*Mus musculus*, representing Rodentia), the house cat (representing Carnivora), and the cow (representing Artiodactyla).

B. The X chromosomes of man and mouse

Many chromosomal segments with the same sequences of two or more homologous loci are syntenic (on the same chromosome) in mouse and man. Gene loci sequences on the X chromosomes of eutherian mammals are especially similar owing to the conserving effect of X inactivation. In the X chromosomes of humans and mice, there are five blocks of gene loci that have the same sequences, although they are in different regions and in part oriented in different directions; these have been retained for about 80 million years, since their divergence from a

common ancestor. (After Lavale and Boyd, 1993).

C. Sequence homologies of the X and the Y chromosomes of man

During the evolution of vertebrates, the X and the Y chromosomes were derived from an autosome. While the Y chromosome has become considerably smaller, the X chromosome has for the most part retained the gene loci of originally autosomal character. In five regions (I–V), the X chromosome and Y chromosome of man contain sequence homologies. The homologous sequences on the extreme distal ends of both chromosomes (I and II) correspond to a pseudo-autosomal region. In region I, homologous pairing and crossing-over occur regularly during meiosis. (After Wolf et al., 1992).

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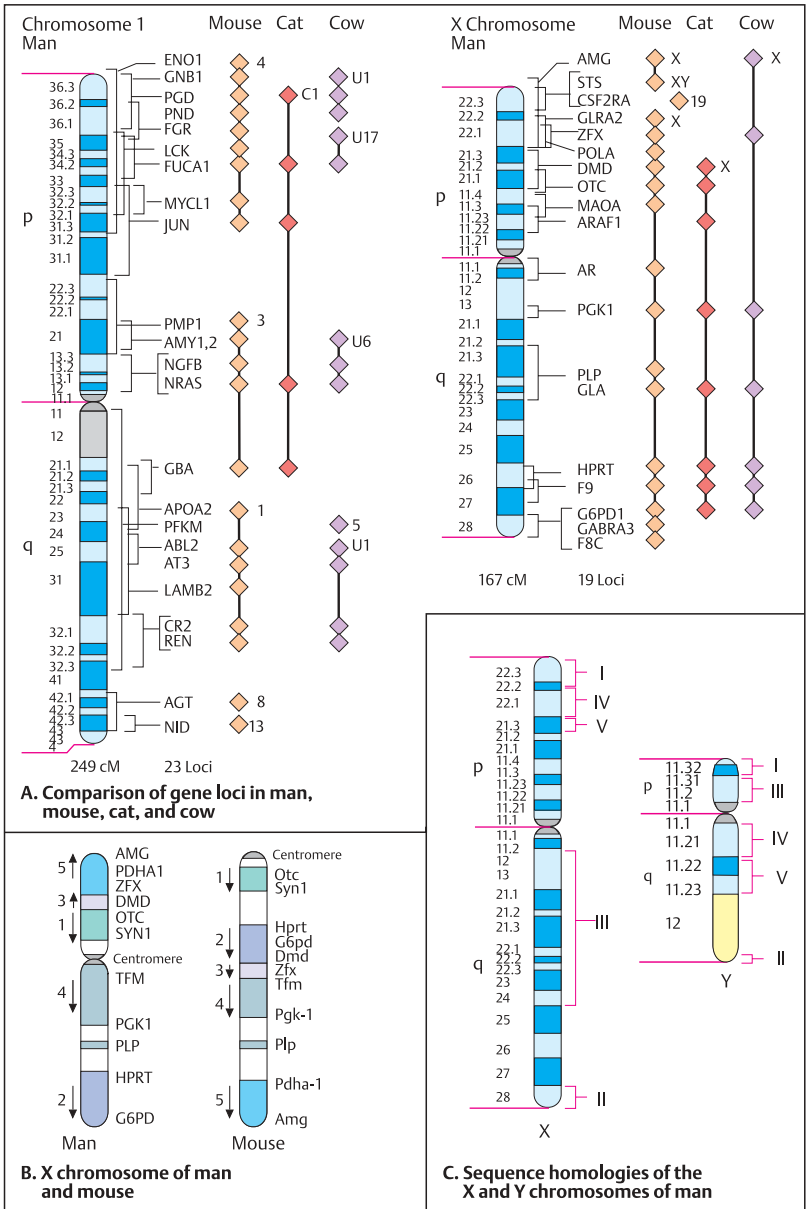
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Comparative genomics: Total number of genes and the “core proteome”

Organism	Number of Genes	Genes Duplicated	Gene Families (Core Proteome)	Type of Organism
<i>H. influenzae</i>	1709	284	1425	Bacterium
<i>C. cerevisiae</i>	6241	1858	4383	Yeast
<i>C. elegans</i>	18424	8971	9453	Nematode
<i>Drosophila</i>	13601	5536	8065	Insect

Data from G. M. Rubin et al.; 2000.



Human Evolution

Human beings belong to the family of Homi-
nidae. Within this family they are a single
living species, *Homo sapiens*. Studies of DNA
sequences obtained from different parts of the
human genome from populations living in
different regions of the world indicate that the
genetic diversity of humans living today is sur-
prisingly limited. The origin of today's humans
can be traced back to about 100 000 to 300 000
years ago in Africa. From here, an anatomically
modern *H. sapiens* ancestor spread out over the
earth and diversified. About 90% of the overall
genetic variation in humans is among individu-
als and only 10% among different ethnic groups.
There are no races of *H. sapiens*; rather, differ-
ent ethnic groups have developed in different
geographic regions during the past 30 000–
40 000 years (see glossary for genetic definition
of race).

A. Hominid family tree with uncertain relationships

The oldest identified hominid skeletal remains
have been found in Eastern Africa. They are at-
tributed to an extinct genus, *Australopithecus*.
Several different species originated about 4.5 to
2 million years ago during the Pliocene epoch
(5.3 to 1.6 million years ago). During this time
fundamental changes in morphology and be-
havior occurred, presumably to adapt to chang-
ing the habitat from the forest to the plains:
bipedalism, a dramatic increase of brain
volume, and anatomic changes in the pharynx
to allow speech were accompanied by tool
making and other complex behavior.
The earliest fossils of *Homo sapiens* are from
100 000 years ago. Modern humans as they
exist today date back about 30 000–40 000
years.

B. Geographic distribution of important hominid finds

Two hypotheses for the origin of modern
humans have been put forward. The first is that
evolution was multiregional, that is, early
humans migrated to different geographic re-
gions, resulting in parallel evolution through-
out the world. The other hypothesis assumes
that all modern humans moved through the
middle east into Europe and Asia between
50 000 and 100 000 years ago and replaced the

descendents of the earlier *Homo erectus*. (Figure
adapted from Wehner and Gehring, 1995).

C. Relationship of modern humans and Neandertals

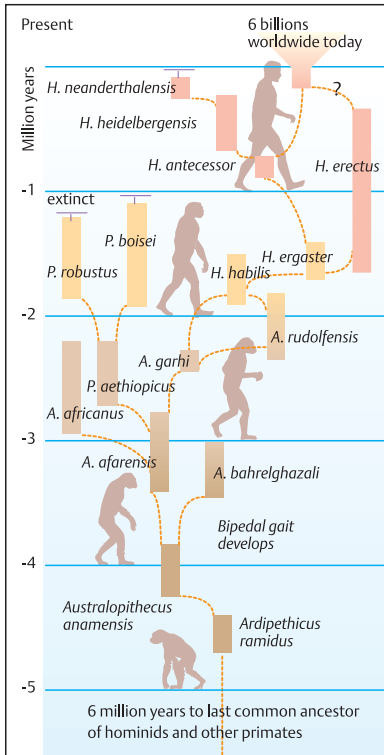
Modern humans and Neandertals coexisted
about 30 000–40 000 years ago. Recent studies
indicate that Neandertals did not contribute
mitochondrial DNA to modern humans. At two
different locations about 2000 km apart,
mtDNA from Neandertal specimens at Feld-
hofer Cave, Neandertal, and Mezmaiskaya Cave
in the northern Caucasus showed only 3.48%
sequence divergence. (Figures adapted from
Krings et al., 1997).

D. Phylogenetic tree reconstruction from mitochondrial DNA

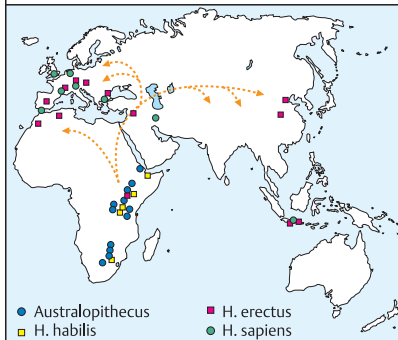
Genetic studies are consistent with the Out of
Africa hypothesis. The strongest evidence came
from an analysis of mitochondrial DNA from 147
modern humans of African, Asian, Australian,
New Guinean, and European origin. The
genealogical tree could be traced back to an an-
cestral haplotype presumably 200 000 years old
("mitochondrial Eve"). (Figure adapted from
Cann et al., 1987).

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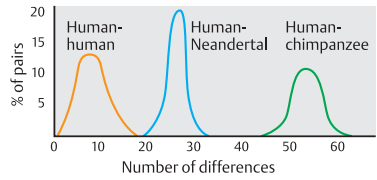
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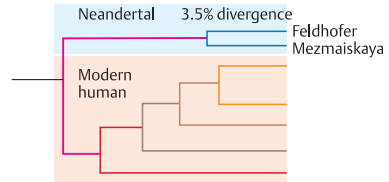
A. Hominid family tree with uncertain relationships



B. Geographic distribution of important hominid finds

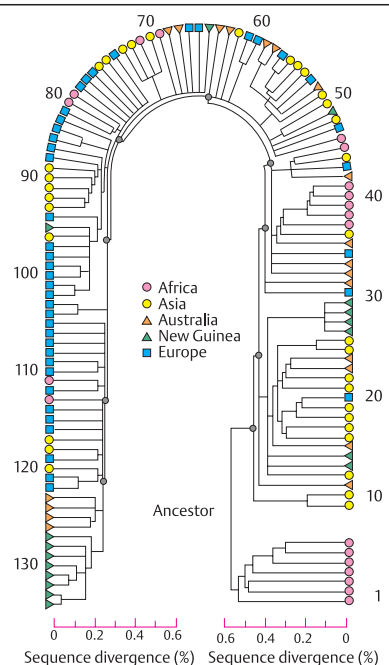


1. Pairwise mtDNA sequence differences



2. Phylogenetic tree

C. Relationship of modern human and Neandertal mtDNA sequence



D. Phylogenetic tree reconstruction of mtDNA evolution in modern humans

Genome Analysis by DNA Microarrays

A microarray or DNA chip is an assembly of oligonucleotides or other DNA probes, e.g., cDNA clones, fixed on a fine grid of surfaces. It is used to analyze the expression states of a series of genes represented in cDNA prepared from mRNA (expression screening) or to recognize sequence variations in genes (screening for DNA variation). The advantages of using microarrays are manifold: simultaneous large-scale analysis of thousands of genes at a time, automation, small sample size, and easy handling given the right equipment. Several manufacturers offer highly efficient microarrays that can accommodate 300 000 DNA probes on a high-density glass slide of small size (e.g. 1.28 cm × 1.28 cm).

Two basic types of DNA microarrays can be distinguished, although many variations are being developed: (i) microarrays of previously prepared DNA clones or PCR products that are attached to the surface, arranged in a high-density gridded array in two-dimensional linear coordinates; (ii) microarrays of oligonucleotides synthesized in situ on a suitable surface. Both types of DNA arrays can be hybridized to labeled DNA probes in solution.

A. Gene expression profile by cDNA array

This figure shows a microarray of 1500 different cDNAs from the human X chromosome. The cDNAs were obtained from lymphoblastoid cells of a normal male (XY) and a normal female (XX). The cDNAs of the male cells were labeled with the fluorochrome Cy3 (green) and the cDNAs of the female cells were labeled with Cy5 (red). The inactivation of most genes in one of the two X chromosomes in female cells leads to a 1 : 1 ratio of cDNAs from the expressed genes in the male and the female X chromosomes (yellow signal at most sites because red fluorescence (female) and green fluorescence (male) signals are superimposed owing to the similar expression levels in male and female cells). An exception is the *XIST* gene, which regulates X inactivation (see p. 228). It is expressed on the inactive X chromosome only. It lights up bright red because it is not expressed on the male X chromosome and is not superimposed by a

green signal. (Photograph courtesy of G. M. Wiczorek, U. Nuber, and H. H. Ropers, Max-Planck Institute for Molecular Genetics, Berlin).

B. Gene expression patterns in human cancer cell lines

Microarrays analyzing the pattern of expression in cancer cells can be expected to have a great impact on diagnosis, surveillance of therapy, and screening for anticancer drugs. Approximately 8000 genes among 60 cell lines derived from different types of cancer have been studied by Ross et al. (2000). A consistent relationship between gene expression patterns and tissue of origin was detectable.

Panel 1 shows the cell-line dendrogram relating the patterns of gene expression with respect to the tissue of origin of the cell lines as derived from 1161 cDNAs in 64 cell lines. Panel 2 shows a colored microarray representation of the data using Cy5-labeled (red) cDNA reverse-transcribed from mRNA isolated from the cell lines compared with Cy3-labeled (green) cDNA derived from reference mRNA. The columns (1161 genes) and the rows (60 cell lines) show in red clusters of increased gene expression at several locations. These observations forecast the future of analysis of altered gene expression patterns in tumor cells. (Figure adapted from Ross, et al. 2000 with kind permission by the authors and Nature Genetics; the data can be visualized at the authors' websites <http://genome-www.stanford.edu/nci60> and <http://discover.nci.nih.gov>).

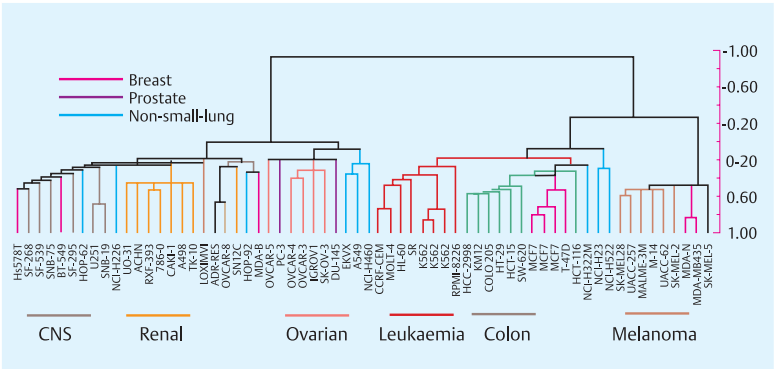
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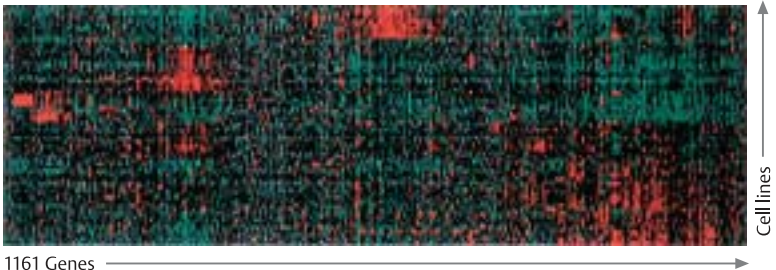
A. Gene expression profile by cDNA array

XIST Expression



1. Cell line dendrogram

- a. Leukaemia cluster b. Epithelial cluster c. Melanoma cluster d. CNS cluster



2. Microarray data

B. Gene expression patterns in human cancer cell lines

Genetics and Medicine

Cell-to-Cell Interactions

Intracellular Signal Transduction Systems

Multicellular organisms depend on communication between cells to assure growth, differentiation, specific functions in different types of cells, and proper response to external stimuli. Specific cell–cell interactions between different types of cells have evolved. A common leitmotif is the specific binding of an extracellular signaling molecule (ligand) to a specific receptor of the target cell to trigger a specific functional response. The vast variety of molecules involved in the many different types of cells can be classified into families of related structure and function (see Lodish et al., 2000; Alberts et al., 1994). Two areas are selected here: the main intracellular functions controlling growth and the receptor tyrosine kinases.

A. Main intracellular functions controlling growth

Growth factors are a large group of different extracellular molecules that bind with high specificity to cell surface receptors (1). Their binding to the receptor (2) activates intracellular signal transduction proteins (3). This initiates a cascade of events resulting in activation of other proteins (often by phosphorylation) that act as second messengers (4). Hormones of different types are a heterogeneous class of signaling molecules (5). They enter the cell either by diffusion through the plasma membrane or by binding to a cell surface receptor (6). Some hormones require an intranuclear receptor (7). Eventually the signal cascade results in activation or inactivation of transcription factors (8). Before transcription and translation ensue, an elaborate system of DNA damage recognition and repair systems (9) make sure that cell proliferation is safe (cell cycle control, 10). In the event that faults in DNA structure have not been repaired prior to replication, an important pathway sacrifices the cell by apoptosis (cell death, 11). (Figure adapted from Lodish et al., 2000.)

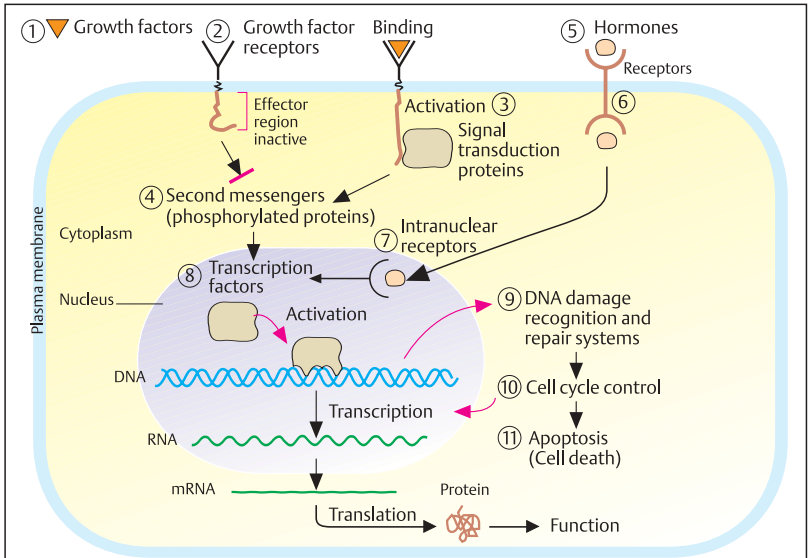
B. Receptor tyrosine kinase family

Like the G protein-coupled receptors (GPCRs, see p. 268) and their effectors, the receptor ty-

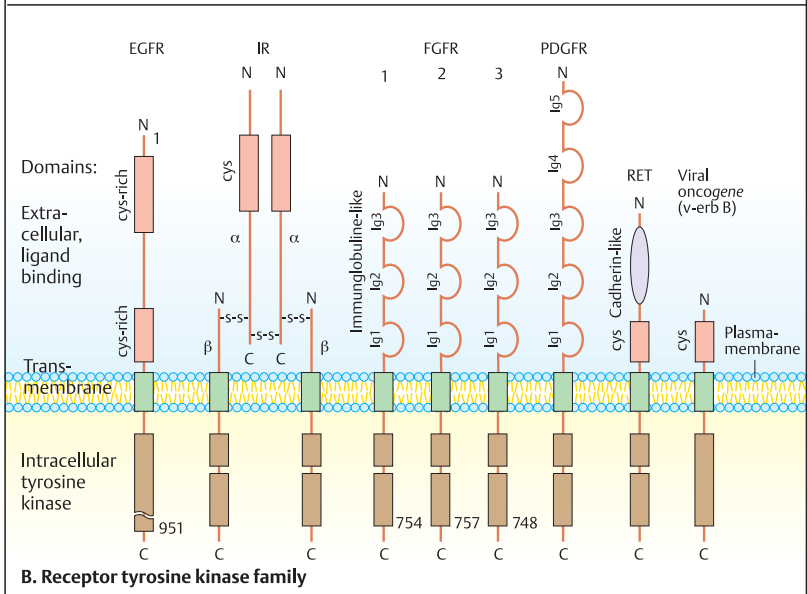
rosine kinases (RTKs) are a major class of cell surface receptors. Their ligands are soluble or membrane-bound growth factor proteins. RTK signaling pathways involve a wide variety of other functions. Mutations in RTKs may send a proliferative signal even in the absence of a growth factor, resulting in errors in embryonic development and differentiation (congenital malformation) or cancer. Of the more than twenty different RTK families, five examples are selected here: the epidermal growth factor receptor (EGFR); insulin receptor (IR); fibroblast growth factor receptor (FGFR) types 1, 2, and 3; platelet-derived growth factor (PDGFR); and RET (rearranged during transformation). These receptors share structural features, although they differ in function. All have a single transmembrane domain and an intracellular tyrosine kinase domain of slightly varied size. The extracellular domains consist of evolutionarily conserved motifs: cysteine-rich regions, immunoglobulin (Ig)-like domains, fibronectin repeats in the tyrosine kinase with Ig and the EGF. RTK mutations cause a group of important human diseases and malformation syndromes. The phenotypes of the mutations differ according to the particular type of RTK involved and the type of mutation.

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A. Main intracellular functions controlling cell growth



B. Receptor tyrosine kinase family

Types of Cell Surface Receptors

Specific receptors on cell surfaces (and in the nucleus or cytosol) convey cell-to-cell signals into the cells and the functional answers. The basic structures of their genes are similar because they have been derived from a relatively small group of ancestral genes. They way they bind to the ligand (the signal-releasing molecule) and the functional answer of the cell are specific. When a ligand binds to a receptor, a series of reactions is initiated that alters the function of the cell. Receptors with direct and indirect ligand effects can be distinguished.

A. Cell surface receptors with direct ligand effect

Many hormones cannot pass through the plasma membrane; instead, they interact with cell surface receptors. Their effects are direct and very rapid. With ligand-activated (or ligand-gated) ion channels (1), binding of the ligand to the receptor changes the conformation of the receptor protein. This causes an ion-specific channel in the receptor protein to open. The resulting flow of ions changes the electric charge of the cell membrane. Receptors with ligand-activated protein kinase (2) further activate a substrate protein. Most protein kinases phosphorylate tyrosine (tyrosine kinase), serine, or threonine by transferring a phosphate residue from adenosine triphosphate (ATP), which is then converted to adenosine diphosphate (ADP). Other receptors mediate the removal of phosphate from a phosphorylated tyrosine side chain by means of their phosphatase activity (3). With one important type of receptor, ligand binding activates guanylate cyclase (4), which catalyzes the formation of cyclic guanosine monophosphate (cGMP) from guanosine triphosphate (GTP). The cGMP functions as a second messenger and brings about a rapid change of activity of enzymes or nonenzymatic proteins. Removal or degradation of the ligand reduces the concentration of the second messenger and ends the reaction. (Diagrams after Lodish et al., 2000.)

B. Hormones with immediate effects on cells

Important examples of hormones that function as ligands are amino acid derivatives, arachidonic acid derivatives, and many peptide hormones.

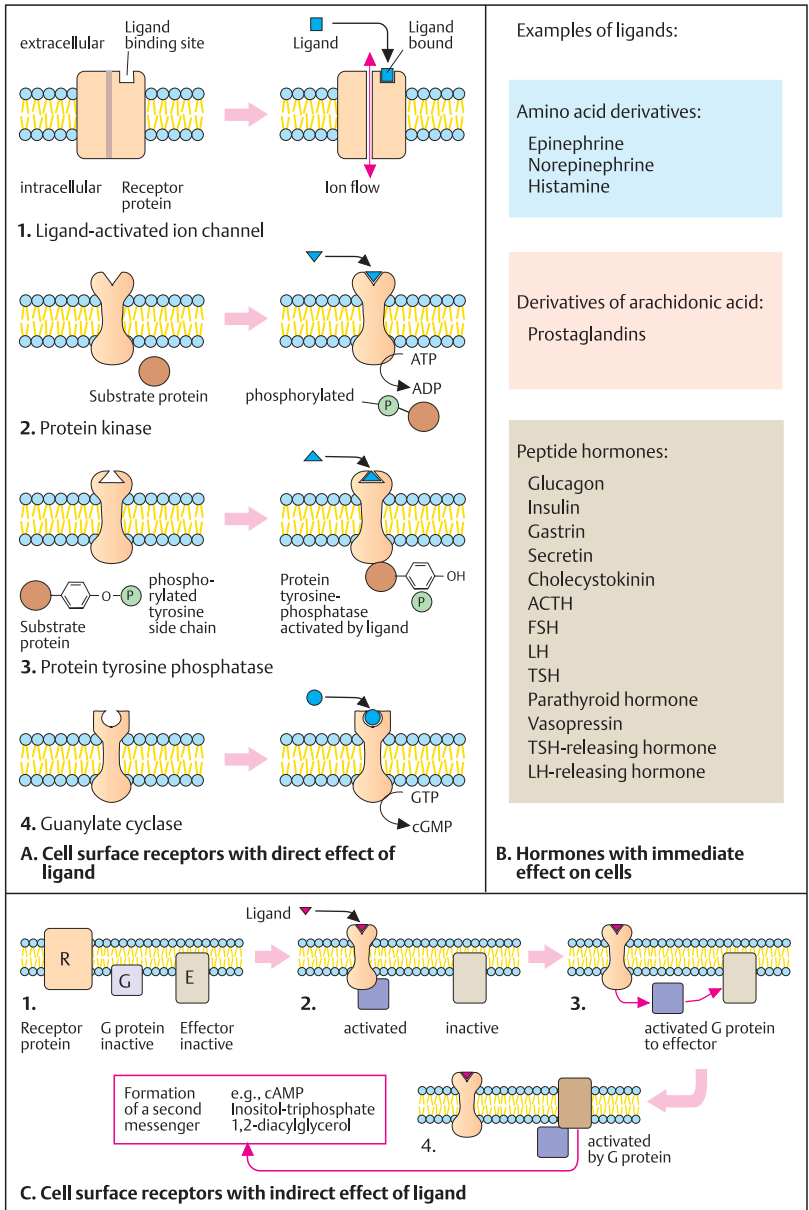
Epinephrine, norepinephrine, and histamine act directly and very rapidly. Peptide hormones such as insulin or adrenocorticotrophic hormone (ACTH) initially occur as precursor polypeptides, which are split by specific proteases to form active molecules. Some peptide hormones are coded for by a common gene; differential RNA splicing of the transcript of this gene produces different precursors for translation. (Abbreviations used: ACTH, adrenocorticotrophic hormone; FSH, follicle-stimulating hormone; LH, leutinizing hormone; TSH, thyroid-stimulating hormone.) (Figure data after Lodish et al., 2000.)

C. Cell surface receptors with indirect ligand effect

Many cell surface receptors act indirectly. When they bind to a ligand they induce a series of intracellular activation steps. This reaction system consists of a receptor protein, a protein (G protein) bound to a guanosine residue, and an enzyme to be activated. Ligand binding alters the receptor protein and activates the G protein (2). This moves to the effector, e.g., an enzyme complex (3), and activates it (4). In this way, a second messenger is formed that triggers further reactions in the cell, e.g., cyclic adenosine monophosphate (cAMP) by means of the enzyme adenylate cyclase (see p. 268).

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G Protein-coupled Receptors

The indirect transmission of signals into the cell is mediated by transmembrane proteins, which traverse the cell membrane. A first messenger, e.g., a hormone like epinephrine, triggers an intracellular reaction by binding to a specific receptor. This leads to activation of a second messenger, which in turn initiates a series of reactions that result in a change of cell function. Many of the genes for the different proteins involved in the indirect transmission of signals are known.

A. Stimulatory G protein (G_s) and hormone–receptor complex

There are many endogenous messengers (hormones) with their own specific receptors. First the hormone binds to the receptor (formation of a hormone–receptor complex). The intracellular transmission of signals is mainly carried out by special guanine-nucleotide-binding proteins, or G proteins. By binding to guanosine triphosphate (GTP, a nucleotide composed of guanine, a sugar, and three phosphate groups), the G protein becomes activated and initiates further reactions. G proteins consist of three subunits: α , β , and γ . The α subunit (stimulatory G protein, G_s) binds to the effector protein. Immediately thereafter, G_s is inactivated (GTPase) by hydrolysis of GTP to GDP (guanosine diphosphate). This transforms the G protein back into an inactive form (G_i). Several human diseases due to defective G protein or a defective G protein receptor are known. (Clapham, 1993.)

B. Four hormone classes

Four principal classes of hormones can be differentiated: (1) amino acid derivatives such as epinephrine and epinephrine derivatives; (2) polypeptides such as glucagon; (3) steroids such as cortisol and its derivatives; and (4) fatty acid derivatives such as the prostaglandins.

C. Formation and hydrolysis of cAMP

The key reaction is the formation of cyclic adenosine monophosphate (cAMP) from adenosine triphosphate (ATP) by means of adenylate cyclase. Intracellular cyclic AMP transmits the activation initiated by the hormone–receptor complex without a molecule having passed through the plasma membrane. cAMP is re-

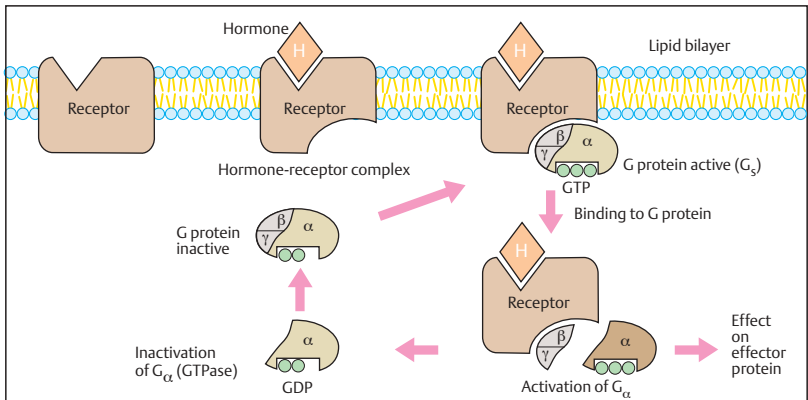
sponsible for many physiological reactions. It becomes inactivated when converted into adenosine monophosphate (AMP) by phosphodiesterase. cGMP (cyclic guanosine monophosphate) functions in the same manner as cAMP to initiate an intracellular reaction.

D. G protein cycle to activate adenylate cyclase

When a hormone binds to its specific receptor, a structural change occurs (1). This activates the α subunit of the G protein, which separates from the β and γ subunits (2). The stimulatory G protein (G_s - α) binds to the effector protein, usually adenylate cyclase, and activates it (3). cAMP is then formed from ATP, while GTP is hydrolyzed to GDP at the G - α subunit. This inactivates the effector protein and the formation of cAMP is terminated. Thus, the signal is of very short duration, and the initial conditions are rapidly restored. Several toxins exert their activity by interrupting this cycle. For example, cholera toxin inhibits inactivation of the G_s - α protein so that adenylate cyclase remains activated and large amounts of sodium and water are lost through the intestinal mucous membranes. (Figures adapted from Watson et al., 1992).

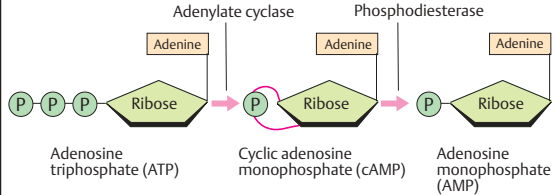
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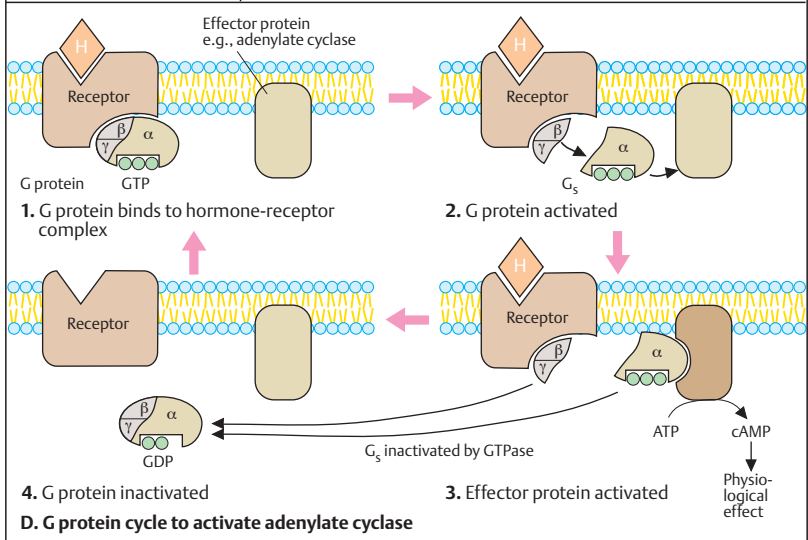
A. Stimulatory G protein (G_s) and hormone-receptor complex

- 1 Norepinephrine (amino acid derivative)
- 2 Glucagon (polypeptide)
- 3 Cortisol (steroid)
- 4 Prostaglandin A_2 (fatty acid derivative)



B. Four hormone types

C. Formation and hydrolysis of cAMP



D. G protein cycle to activate adenylate cyclase

Transmembrane Signal Transmitters

Functional signals between cells are received by transmembrane proteins as signal transmitters. During evolution, relatively simple precursor genes for such proteins gave rise to multiple structurally and functionally related genes. Their corresponding proteins serve to transmit ions (sodium, potassium, calcium, chloride, and others), as neurotransmitters, and for perception of light and odors, etc. Cloning of these genes has yielded insight into the variety of functions of transmembrane signal transmitters. Their general structure can be traced back to an evolutionarily conserved ancestral molecule.

A. Transmembrane structure of voltage-gated ion channels

The direct flow of ions across the cell membrane is regulated by ion channels. The transmembrane proteins, composed of several domains, are arranged so that they form pores that can be opened and closed. The simplest model is the potassium channel (1). This membrane-bound polypeptide contains six transmembrane domains. The amino and the carboxy ends of the protein lie within the cell. Changes in cell membrane potential or voltage cause the channel to open (or close) in order to initiate (or terminate) a brief flow of ions. Domain 4, which is composed of polar amino acids, is crucial for the flow of ions. Sodium and calcium ion channels consist of four subunits (2) of similar structure, each resembling a potassium channel. The similarity is due to the common evolutionary origin of their genes. The four subunits of the sodium channel (3) are positioned to form a very narrow porelike passage, much narrower than a potassium channel, through the plasma membrane. Ion transport is brought about by membrane depolarization (3 and 4). (Figure after Watson et al., 1992.)

B. Seven-helix structure of transmembrane signal transmitters

Indirect transmission of signals is more frequent than the direct transport of ions or ligand-gated impulse transmission. Here, the transmembrane protein is involved only in the first step of signal transmission. Further steps

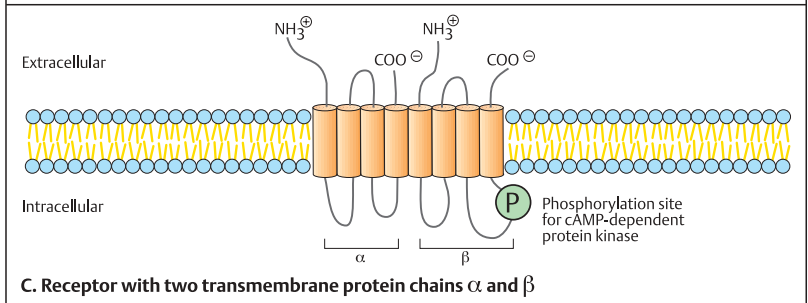
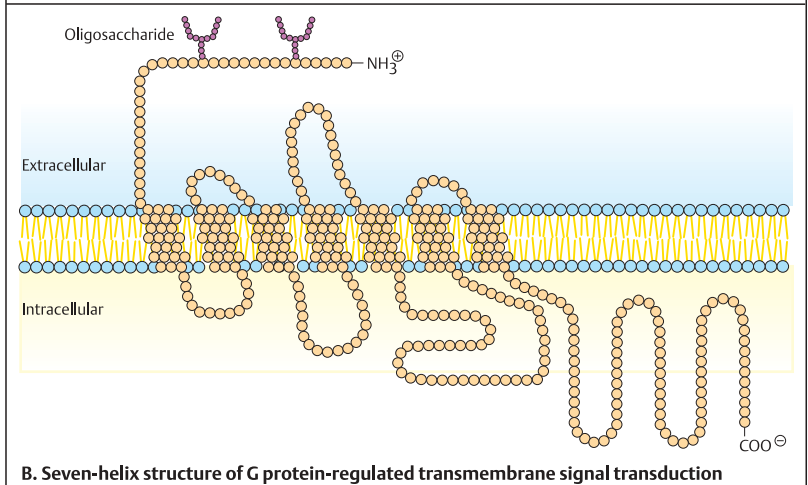
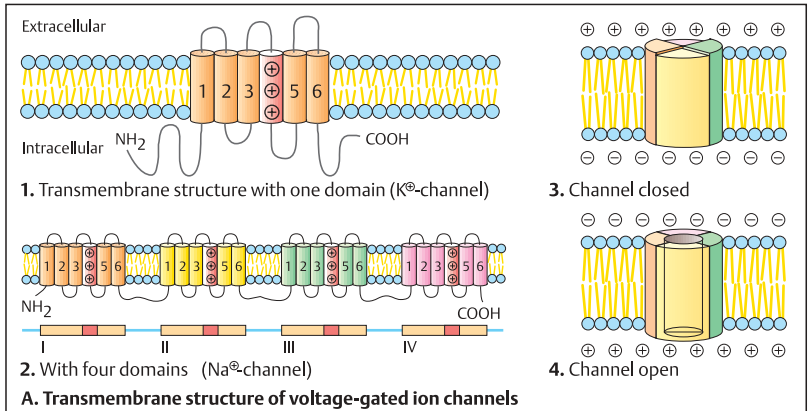
follow. An especially common structural motif is a transmembrane protein containing seven α helices within the plasma membrane. The amino end is extracellular; the carboxy end is intracellular. Different oligosaccharide side chains are usually bound to the extracellular domains. The intracellular domains have binding sites for other molecules involved in signal transmission. The seven-helix motif is the characteristic structure of G protein-binding receptors (p. 268). As the G proteins themselves, these receptors and their genes form a large family with a long evolutionary history. In yeast, they serve to discern the pheromones of the mating types (p. 186); in higher organisms they are the basis for transmitting signals of vision, smell, and taste (p. 278–286). (Figure redrawn from Stryer, 1995).

C. A receptor with two transmembrane protein chains, α and β

The receptor for γ -aminobutyric acid (GABA) utilizes two transmembrane protein subunits, α and β . Both the amino and the carboxy ends are extracellular. The two chains are coded for by different genes. Several oligosaccharide side chains are present on the extracellular side. The β chain contains a phosphorylation site for cAMP-dependent protein kinase. (Figures adapted from Watson et al., 1992).

References

- Sabatini, D.D., Adesnik, M.B.: The biogenesis of membranes and organelles. pp. 459–553, In: Sriver, C.R., et al., eds., *The Metabolic and Molecular Bases of Inherited Disease*. 7th ed. McGraw-Hill, New York, 1995.
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Receptors of Neurotransmitters

Impulses are relayed between nerve cells or between nerve and muscle cells by various transmitter molecules (neurotransmitters). Their effects are further relayed by receptors in the cell membrane. Receptors can be differentiated according to their structure, which in turn determines their specificity.

A. Acetylcholine as a neurotransmitter

Cholinergic synapses convey the nerve impulse from one nerve cell to another or from a nerve cell to a muscle cell (motor endplate). Acetylcholine leads to postsynaptic depolarization through the release of potassium ions (K^+) and the uptake of sodium ions (Na^+). This process is regulated by an acetylcholine receptor.

B. Acetylcholine receptors

The acetylcholine receptors are of two genetically and functionally different types. Pharmacologically they can be differentiated according to the effects of nicotine and muscarine. The nicotine-sensitive acetylcholine receptor is an ion channel for potassium and sodium. It consists of five subunits: two α , one β , one γ , and one δ (1). Acetylcholine binds as a ligand to the two α subunits. Each subunit consists of four transmembrane domains (2). Each subunit is encoded by its own gene (3). These genes have similar structures and nucleotide base sequences. The ligand-gated ion channel is an example of direct transport without an intermediate carrier. A mutation in the second transmembrane region has been shown to change the ion selectivity from cations to anions (Galzi, 1992.)

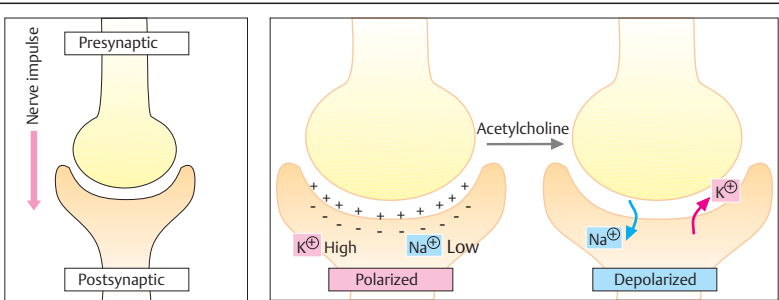
The muscarine-sensitive type of acetylcholine receptor is a protein that contains seven transmembrane subunits (4). Since each exists in the form of an α helix, it is referred to as a seven-helix transmembrane protein (p. 270). The amino end (NH_2) lies extracellularly; the carboxy end ($COOH$), intracellularly. The transmembrane domains are connected by intracellular and extracellular polypeptide loops (4). Different domains of the whole protein are distinguished (5) according to location and the relative proportion of hydrophilic and hydrophobic amino acids. The amino end and the carboxy end each form a domain just like the intracellular (a-c), and extracellular portions (d-f).

The transmembrane domains located within the plasma membrane (1–7) consist for the most part of hydrophobic amino acids. The structure of the gene product corresponds to the general structure of the gene (6). The different domains are coded for by individual exons. The DNA nucleotide sequences within functionally similar domains are similar.

The seven-helix transmembrane motif occurs in many receptors. The general structures of the genes and of the gene products are very similar, but they differ in their specificity of binding to other functionally relevant molecules (G proteins). They play a role not only as neurotransmitters but also in the transmission of light, odors, and taste. (Figures based on Watson et al., 1992.)

References

- Galzi, G.L.: Mutations in the channel domain of a neuronal nicotinic receptor convert ion selectivity from cationic to anionic. *Nature* 359:500–505, 1992.
- Watson, J.D., Gilman, M., Witkowski, J., Zoller, M.: *Recombinant DNA*. 2nd ed. W.H. Freeman, Scientific American Books, New York, 1992.



Cholinergic synapse (nerve/nerve or nerve/muscle)

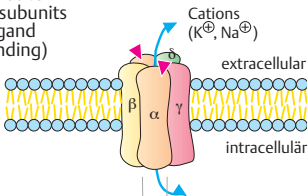
A. Acetylcholine as neurotransmitter

Two types of acetylcholine receptors

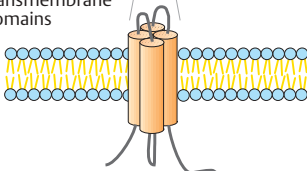
Cation-specific channel
in muscle of vertebrates
(nicotine-sensitive)

Seven-helix transmembrane protein
bound to G proteins
(muscarine-sensitive)

1. Acetylcholine binds to α -subunits (ligand binding)



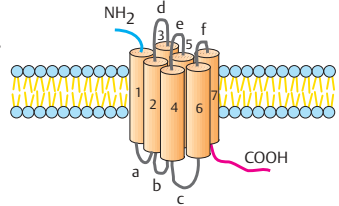
2. Each subunit has four transmembrane domains



3. One gene for each subunit:

2 for α -subunits
1 for β -subunit
1 for γ -subunit
1 for δ -subunit

- 4.

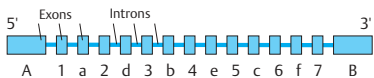


- 5.



A Amino end
B Carboxy end
a - c intracellular domains
d - f extracellular domains
1 - 7 transmembrane domains (hydrophobic)

6. Gene structure (diagram)



The different domains are encoded by individual exons

B. Acetylcholine receptor

Genetic Defects in Ion Channels

More than 20 different disorders due to defective ion channel proteins resulting from gene mutations are known. Such disorders include cystic fibrosis (see p.276), the long-QT syndrome, a special type of deafness, hereditary hypertension (Liddle syndrome), familial persistent hyperinsulinemic hypoglycemia of infancy, some hereditary muscle diseases, and malignant hyperthermia (see p.372), among other disorders.

A. Long-QT syndrome, a genetic cardiac arrhythmia

Congenital long-QT syndrome is characterized by a prolonged QT interval in the electrocardiogram (more than 460 ms, corrected for heart rate), sudden attacks of missed heart beats (syncope) or series of rapid heart beats (torsades de pointes), and an increased risk for sudden death from ventricular fibrillation in children and young adults.

B. Different molecular types of long-QT syndrome

Prolongation of the QT interval in the electrocardiogram results from an increase in the duration of the cardiac action potential (1). The normal potential lasts about 300 ms (phases 1 and 2). The resting membrane potential (phase 3) is reached by progressive inactivation of calcium currents and increasing depletion of potassium currents, which repolarize the cell. In phase 0 the cell is quickly depolarized by acti-

vated sodium currents following an excitatory stimulus. LQT1 accounts for about half of the patients with long-QT syndrome. The gene for LQT2 encodes a 1195-amino-acid transmembrane protein responsible for the other major potassium channel that participates in phase 3 repolarization (*HERG* stands for (human-ether-r-go-related gene, a *Drosophila* homologue). LQT3, a sodium channel protein, consists of four subunits, each containing six transmembrane domains and a number of phosphate-binding sites. Homozygosity for LQT1 (*KVLQT1* gene) or LQT5 (*KCNE1* gene) causes a form of long-QT syndrome associated with deafness, the Jervell and Lange-Nielsen syndrome. (Figure adapted from Ackerman and Clapham, 1997.) It is important to distinguish the different types because the choice of medication differs.

References

Ackerman, M.J., Clapham, D.D.: Ion channels—Basic science and clinical disease. New Eng. J. Med. **336**: 1575 – 1586, 1997.
Keating, M.T., Sanguinetti, M.C.: Molecular genetic insights into cardiovascular disease. Science **272**:681 – 685, 1996.
Schulze-Bahr, E., et al.: *KCNE1* mutations cause Jervell and Lange-Nielsen syndrome. Nature Genet. **17**:267 – 268, 1997.
Schulze-Bahr, E., et al.: The long-QT syndrome. Current status of molecular mechanisms. Z. Kardiol. **88**:245 – 254, 1999.
Viskin, S.: Long QT syndromes and torsades de pointes. Lancet **354**:1625 – 1633, 1999.

Examples of Diseases due to Genetic Ion Channel Defects

Disease	Inheritance	Ion Channel Gene	Locus
Cystic fibrosis	AR	<i>CFTR</i> (chloride)	7q31
Long-QT syndrome (6 types)	AD	<i>KVLQT1</i> (cardiac potassium)	11p15.5
		<i>HERG</i>	7q35 – 36
		<i>SCNA5</i> (cardiac sodium)	3p21
		Three other types (see text)	
Malignant hyperthermia	AD	(Ryanodine, muscle calcium)	19q13.1
		(skeletal-muscle sodium)	17q23 – 25

AD, autosomal dominant; AR, autosomal recessive. (About 18 others not listed, adapted from Ackerman and Clapham, 1997.)

Prolonged QT interval in the electrocardiogram

Syncope

Sudden death

Autosomal dominant

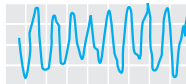
Six genes involved (LQT1 - LQT6)



Prolonged QT



Torsade de pointes



Romano Ward syndrome (Long-QT syndrome)

Type	Locus	Gene
LQT1	11p15.5	<i>KCNQ1 (KVLQT1)</i>
LQT2	7q35-36	<i>HERG</i>
LQT3	3p21-24	<i>SCNA5</i>
LQT4	4q25-27	unknown
LQT5	21q22.1	<i>KCNE1</i>
LQT6	21q21.1	<i>KCNE2</i>

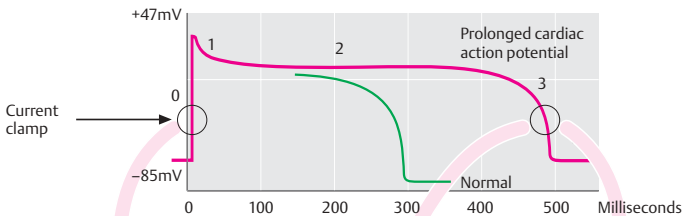
Long-QT and Deafness (Jervell and Lange-Nielsen) due to allelic mutations at LQT1 and LQT5 (autosomal recessive)

1. Main features

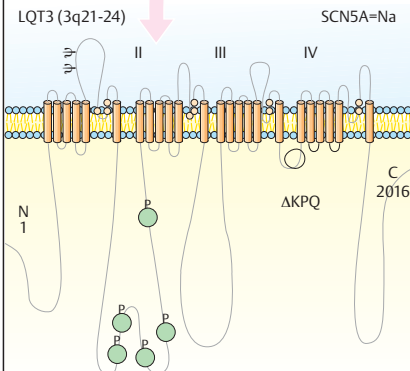
2. Electrocardiogram

3. Genetics

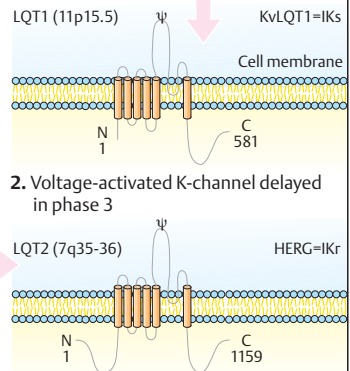
A. Long-QT syndrome, a genetic cardiac arrhythmia



1. Increased duration of cardiac action potential



4. Na-channel fails to inactivate completely during phase 0



2. Voltage-activated K-channel delayed in phase 3

3. Voltage-gated K-channel delayed in phase 3

B. Different molecular types of long-QT syndrome

Chloride Channel Defects: Cystic Fibrosis

Cystic fibrosis (mucoviscidosis) is a highly variable multisystemic disorder due to mutations of the cystic fibrosis transmembrane conduction regulator gene (*CFTR*). Cystic fibrosis (CF) is one of the most frequent autosomal recessive hereditary diseases in populations of European origin (about 1 in 2500 newborns). The high frequency of heterozygotes (1:25) is thought to result from their selective advantage due to reduced liability to epidemic diarrhea (cholera).

A. Cystic fibrosis: clinical aspects

The disease primarily affects the bronchial system and the gastrointestinal tract. Viscous mucus formation leading to frequent, recurrent bronchopulmonic infections and eventually chronic oxygen deficiency characterize the common, severe form of the disease. The average life expectancy in typical CF is about 30 years. The disease may take a less severe, almost mild course. Congenital bilateral absence of the vas deferens (CBAVD) occurs in 95% of patients with CF. It may be the only manifestation in individuals with different mutant allelic combinations at the CF locus.

B. Positional cloning of the gene for cystic fibrosis (CF)

The *CFTR* gene was isolated on the basis of its chromosomal location (positional cloning) on the long arm of chromosome 7 at q31 (7q31). Since the gene could be mapped to the long arm of chromosome 7 near a marker locus D7S15, a long-range restriction map comprising about 1500 kb containing the presumptive CF locus flanked by two marker loci, MET and D7S8, was constructed. From there a region of 250 kb was isolated by a combination of chromosome walking and chromosome jumping. Several genes were located in this region (candidate genes) between the marker loci D7S340 and D7S424. The gene sought was identified by the finding of mutations of this gene in patients but not in controls, by comparing with similar genes in other organisms (evolutionary conservation), by determining its exon/intron structure, by sequencing it, and by determining the expression pattern of the gene in different tissues.

C. The *CFTR* gene and its protein

The *CFTR* gene is large, extending over 250 kb of genomic DNA, and is organized into 27 exons (24 are shown in the diagram) encoding a 6.5 kb transcript with several alternatively spliced forms of mRNA. The *CFTR* protein has 1480 amino acids. It is a membrane-bound chloride ion channel regulator with several functional domains: two nucleotide-binding domains (encoded by exons 9–12 and 19–23), a regulatory domain (exons 12–14a), and two transmembrane-spanning domains (exons 3–7 and 14b–18). Each of the two transmembrane regions consists of six transmembrane segments. The nucleotide-binding domain 1 (NBD1) confers cAMP-regulated chloride channel activity. The most common mutation (occurring in 66% of patients), a deletion of a phenylalanine codon in position 508 ($\Delta F508$), is located here. The protein is a member of the ATP-binding cassette (ABC) family of transporters. The R domain contains putative sites for protein kinase A and protein kinase C phosphorylation. *CFTR* is widely expressed in epithelial cells. The more than 800 different mutations observed in the *CFTR* gene (see <http://www.genet.sickkids.on.ca/cftr>) can be grouped into five different functional classes: (i) abolished synthesis of full-length protein, (ii) block in protein processing, (iii) reduced chloride channel regulation, (iv) reduced chloride channel conductance, (v) reduced amount of normal *CFTR* protein. The underlying genetic defects include missense mutations, nonsense mutations, RNA splicing mutations, and deletions. Aside from $\Delta F508$ in about 66% of patients, the most frequent mutations worldwide are G542X (2.4%), G551D (1.6%), N1303K (1.3%), and W1282X (1.2%).

References

- Chillon, M., et al.: Mutations in the cystic fibrosis gene in patients with congenital absence of the vas deferens. *N. Engl. J. Med.* **332**:1475–1480, 1995.
- Rosenbluth, D.B., Brody, S.L.: Cystic fibrosis, pp. 329–338, In: J.L. Jameson, ed., *Principles of Molecular Medicine*. Human Press, Totowa, N.J., 1998.
- Rosenstein, B.J., Zeitline, P.C.: Cystic fibrosis. *Lancet* **351**:277–282, 1998.
- Tsui, L.C.: The spectrum of cystic fibrosis mutations. *Trends Genet.* **8**:392–398, 1992.

Cystic fibrosis (Mucoviscidosis)

Severe progressive disease of the bronchial system and gastro-intestinal tract

Disturbed function of a chloride ion channel by mutations of the *CFTR* gene

Autosomal recessive

Gene locus 7q31.3

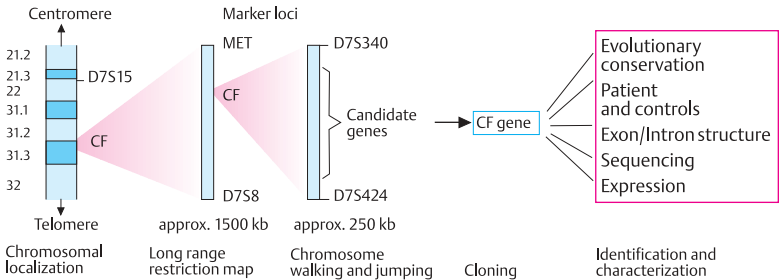
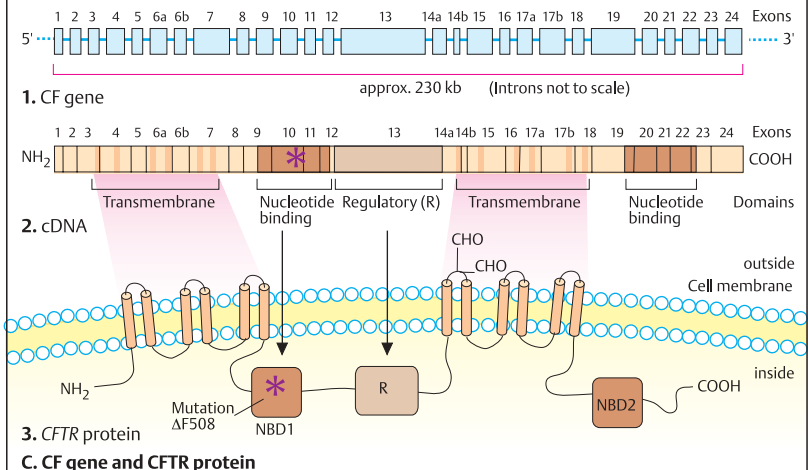
Disease incidence approx. 1:2500

Heterozygote frequency approx. 1:25

Mutation $\Delta F508$ in approx. 70%

**A. Cystic fibrosis, a very frequent recessive disease in Europe and North America**

Part of chromosome 7

**B. Positional cloning of the gene for cystic fibrosis (CF)**

Rhodopsin, a Photoreceptor

The human retina contains about 110 million rod cells for vision in the dark and 6 million cone cells for color vision in the light. These cells contain photoreceptors that convert light into a nerve impulse. Rhodopsin is the photoreceptor for weak light. The light-transmitting system consists of numerous components coded for by genes that are similar in structure and function to genes for other transmembrane signal-transmitting molecules.

A. Rod cells

A rod cell consists of an outer segment with a photoreceptor region and an inner segment comprising cell nucleus and cytoplasm with endoplasmic reticulum, Golgi apparatus, and mitochondria. The outer segment contains about 1000 disks with rhodopsin molecules in the membrane. In the periphery, the approximately 16 nm thick disks are folded by the protein peripherin. (Diagram after Stryer, 1995).

B. Photoactivation

In 1958, George Wald and co-workers discovered that light isomerizes 11-*cis*-retinal (1) very rapidly into all-*trans*-retinal (2), a form that practically does not exist in the dark (~1 molecule/1000 years). The light-induced structural change is so great that the resulting atomic motion can trigger a reliable and reproducible nerve impulse. The absorption spectrum of rhodopsin (3) corresponds to the spectrum of sunlight, with an optimum at a wavelength of 500 nm. Although vertebrates, arthropods, and mollusks have anatomically quite different types of eyes, all three phyla use 11-*cis*-retinal for photoactivation.

C. Light cascade

Photoactivated rhodopsin triggers a series of enzymatic steps (light cascade). First, a signal-transmitting protein of visualization, transducin, is activated by photoactivated rhodopsin. Transducin belongs to the G protein family, i.e., it can assume an inactive GDP and an active GTP form. GTP activates phosphodiesterase. This very rapidly hydrolyzes cGMP and lowers the cGMP concentration in cytosol, which leads to closure of the sodium ion channels. Immediately thereafter, phosphodiesterase is inactivated by means of a G protein cycle.

D. Rhodopsin

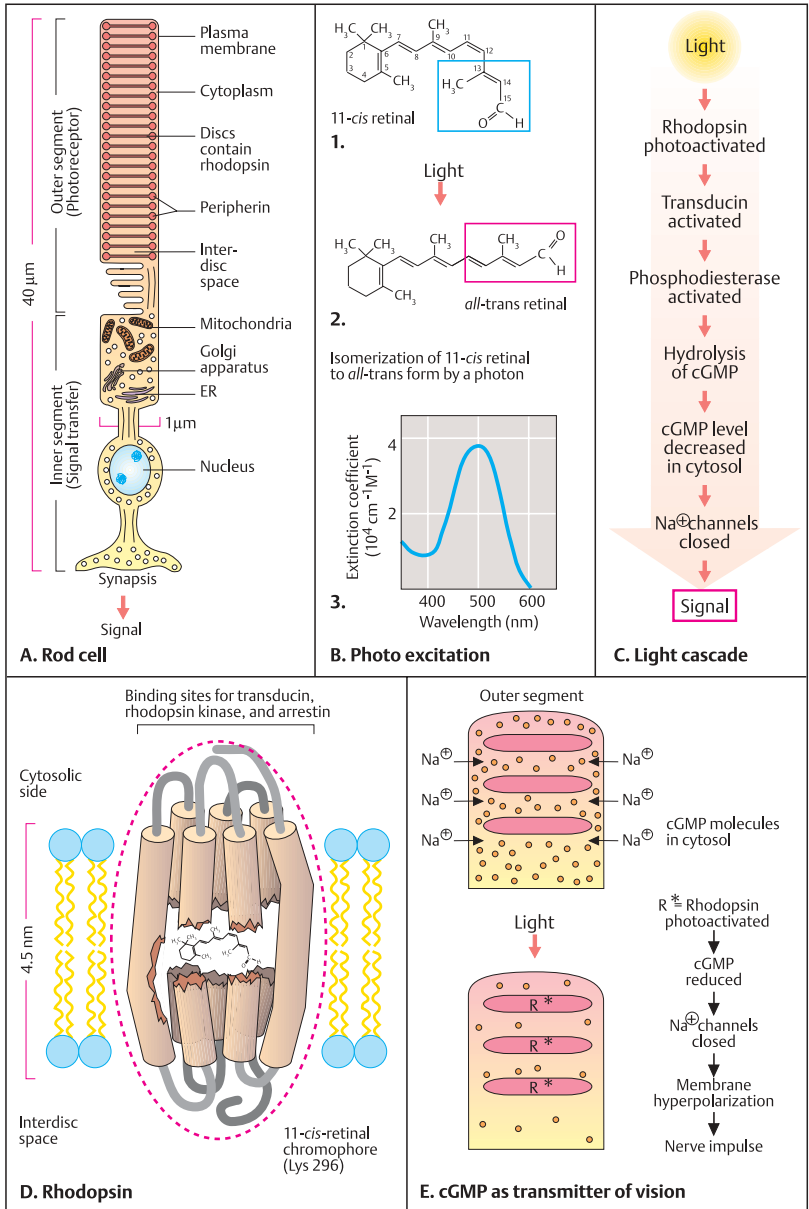
Rhodopsin is a seven-helix transmembrane protein with binding sites for functionally important molecules such as transducin, rhodopsin kinase, and arrestin on the cytosol side. The binding site of the light-sensitive molecule (chromophore) is lysine in position 296 of the seventh transmembrane domain. The light-absorbing group consists of 11-*cis*-retinal. The amino end of rhodopsin is located in the disk interspaces, and the carboxy end on the cytosol side. About half of the molecule is contained in the seven transmembrane hydrophobic domains, one-fourth in the disk interspaces and one-fourth on the cytosol side.

E. cGMP as transmitter in the visualization process

The light cascade ends with rapid hydrolysis of cGMP, the internal transmitter in visualization. This leads to rapid closure of the sodium ion channels and hyperpolarization of the membrane to initiate nerve impulse, which is transmitted as a signal to the brain.

References

- Palczewski, K. et al.: Crystal structure of rhodopsin: A G protein-coupled receptor. *Science* **289**:739–745, 2000.
- Schoenlein, R.W., et al.: The first step in vision: Femtosecond isomerization of rhodopsin. *Science* **254**:412–415, 1991.
- Stryer, L.: *Biochemistry*. 4th ed. W.H. Freeman, New York, 1995.
- Stryer, L.: Molecular basis of visual excitation. *Cold Spring Harbor Symp Quant Biol.* **53**:28–294, 1988.



Mutations in Rhodopsin

Retinitis pigmentosa (RP) is a genetically heterogeneous group of diseases that lead to pigmental degeneration of the retina and progressive blindness. Numerous mutations in the rhodopsin gene have been shown to be the cause of different forms of RP. Mutations in other genes coding for proteins of the light cascade may also cause retinitis pigmentosa.

A. Retinitis pigmentosa

The fundus of the eye shows distinct displacement of pigmentation, with irregular hyperpigmentation and depigmentation. The papilla (optic disk) shows waxy yellow discoloration. The loss of vision, especially in dim light (night blindness), proceeds from the periphery to the center at different rates depending on the form of the disease, until only a very narrow central visual field remains. (Photograph from E. Zrenner, Tübingen.)

B. Point mutation in codon 23

The first point mutation demonstrated in the rhodopsin gene (Dryja et al., 1990) was a transversion from cytosine to adenine in codon 23. This changed the codon CCC for proline (Pro) into CAC for histidine (His). Since the proline in position 23 occurs in more than ten related G protein receptors, it must be very important for normal function.

C. Mutations in rhodopsin

The gene locus for rhodopsin (RHO) in man lies on the long arm of chromosome 3 in region 2, band 1.4 (3q21.4). Dominant and autosomal recessive inherited mutations have been demonstrated in humans. Most mutations lead to the exchange of an amino acid, although deletions may also occur. Of the 348 amino acids of rhodopsin, 38 are identical (invariant) at various positions in vertebrates. More than 100 different mutations are known for autosomal dominant inherited RP. An increasing number of mutations are recognized to cause autosomal recessive RP. In addition, mutations in several other gene loci have been recognized to lead to retinitis pigmentosa, e.g. mutations in the gene for peripherin on the short arm of chromosome 6 in humans (6p) and a locus in the centromeric region of chromosome 8. Other photoreceptor gene disease loci are the α and β subunits of phosphodiesterase (PDE).

D. Demonstration of a mutation in codon 23 by means of oligonucleotides after PCR

This pedigree (1) with autosomal dominant inherited retinitis pigmentosa due to mutation in codon 23 includes 13 affected individuals in three generations (affected females, black circles; affected males, black squares). Using polymerase chain reaction (PCR) (see p. 166), Dryja et al. (1990) demonstrated the mutation in amplified fragments of exon 1 (2). The normal oligonucleotide corresponds to the normal sequence between codons 26 and 20. The mutant sequence of the oligomere RP contains the mutant sequence CAC. All affected individuals gave a hybridization signal with the RP oligomer (2) (II-2, II-12, and III-4 were not examined), whereas unaffected individuals did not (see p. 408 for demonstration of a point mutation with oligonucleotides).

References

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- Dryja, T.P.: Retinitis pigmentosa, pp. 4297–4309, In: C.R. Scriver, et al., eds., *The Metabolic and Molecular Bases of Inherited Disease*. 7th ed. McGraw-Hill, New York, 1995.
- Dryja, T.P., et al.: A point mutation of the rhodopsin gene in one form of retinitis pigmentosa. *Nature* **343**:364–366, 1990.
- McInnes, R.R., Bascom, R.A.: Retinal genetics: a nullifying effect for rhodopsin. *Nature Genetics* **1**:155–157, 1992.
- Wright, A.F.: New insights into genetic eye disease. *Trends Genet.* **8**:85–91, 1992.

A group of hereditary diseases with degeneration of the retinal pigment

Night blindness

Progressive loss of vision

Frequency about 1:3500



Typical fundus with pigment changes, narrow vessels, and pale, waxy optical nerve

Frequency of the different genetic forms

25% autosomal dominant
20% autosomal recessive
8% X-chromosomal
47% Mode of inheritance uncertain in an individual patient

Important diagnostic signs

Fundus:
narrow vessels
pale optic nerve
macular changes
widened light reflex
pigment epithelium changes
electroretinogram silent

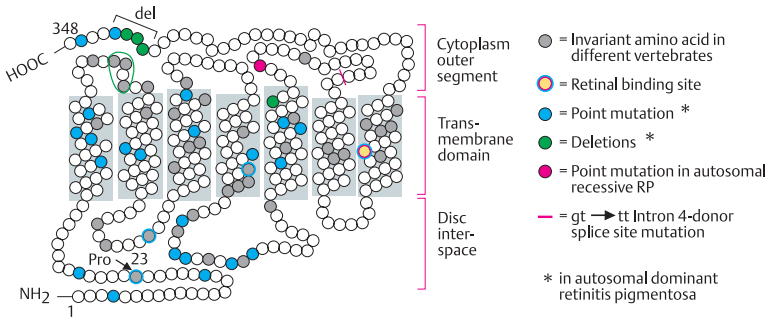
Secondary changes in the anterior chamber:
vitreous body changes

Cataract
Myopia

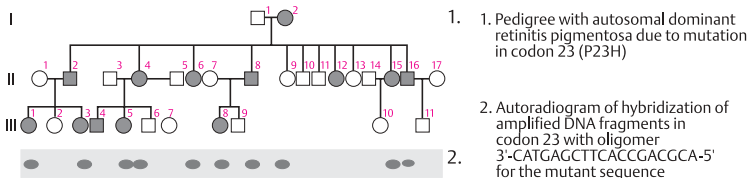
normal	C	T	A	G		mutant	C	T	A	G
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	A	-	-	-	-	-
-	-	-	-	-	T	-	-	-	-	-
-	-	-	-	-	G	-	-	-	-	-
-	-	-	-	-	A	-	-	-	-	-
-	-	-	-	-	G	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	T	-	-	-	-	-
-	-	-	-	-	T	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	G	-	-	-	-	-
-	-	-	-	-	A	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	G	-	-	-	-	-
-	-	-	-	-	C	-	-	-	-	-
-	-	-	-	-	A	-	-	-	-	-

A. Retinitis pigmentosa

B. Point mutations in codon 23



C. Mutations in rhodopsin



D. Demonstration of mutation P23H in codon 23 by oligonucleotides after PCR

Color Vision

As suggested by Thomas Young in 1802, color vision in humans is mediated by three receptor types in the cone cells of the retina, one each for blue, green, and red.

A. Genes for photoreceptor proteins in cones

The gene for the blue receptor is autosomal; the genes for the red and green receptors are X chromosomal. The absorption spectra of the three receptors show maxima of 426 nm for blue, about 530 for green, and about 550 for red. The red receptor was discovered to be polymorphic, with two somewhat different absorption maxima at 552 and 557 nm.

B. Evolution of the genes for visual pigment photoreceptors

The photoreceptor genes arose from a single ancestral gene (protogene). The rhodopsin-transducin pair is found in invertebrates and is at least 700 million years old. The blue receptor is almost as old as rhodopsin, about 500 million years. The separation into a receptor for green and one for red must have occurred only about 30 million years ago, after the Old World and New World apes separated, since man and the Old World apes have three cone pigments whereas New World apes have two.

C. Structural similarity of the visual pigments

In 1986, J. Nathans and co-workers sequenced the genes for color photoreceptors and observed marked structural similarities, especially of the green and red receptor genes. Here the gene products (the receptors) are shown and their similarities are compared. The dark dots indicate variant amino acids; the light dots are identical amino acids, given in percentages.

D. Polymorphism in the photoreceptor for red

A. G. Motulsky and co-workers (Winderickx et al., 1992) demonstrated variant codons in three regions of the red receptor gene (1). Serine was found at position 180 in 60% of the investigated males; alanine in 40%. Position 230 showed polymorphism of isoleucine (Ile) and threonine (Thr); position 233 of alanine (Ala) and serine

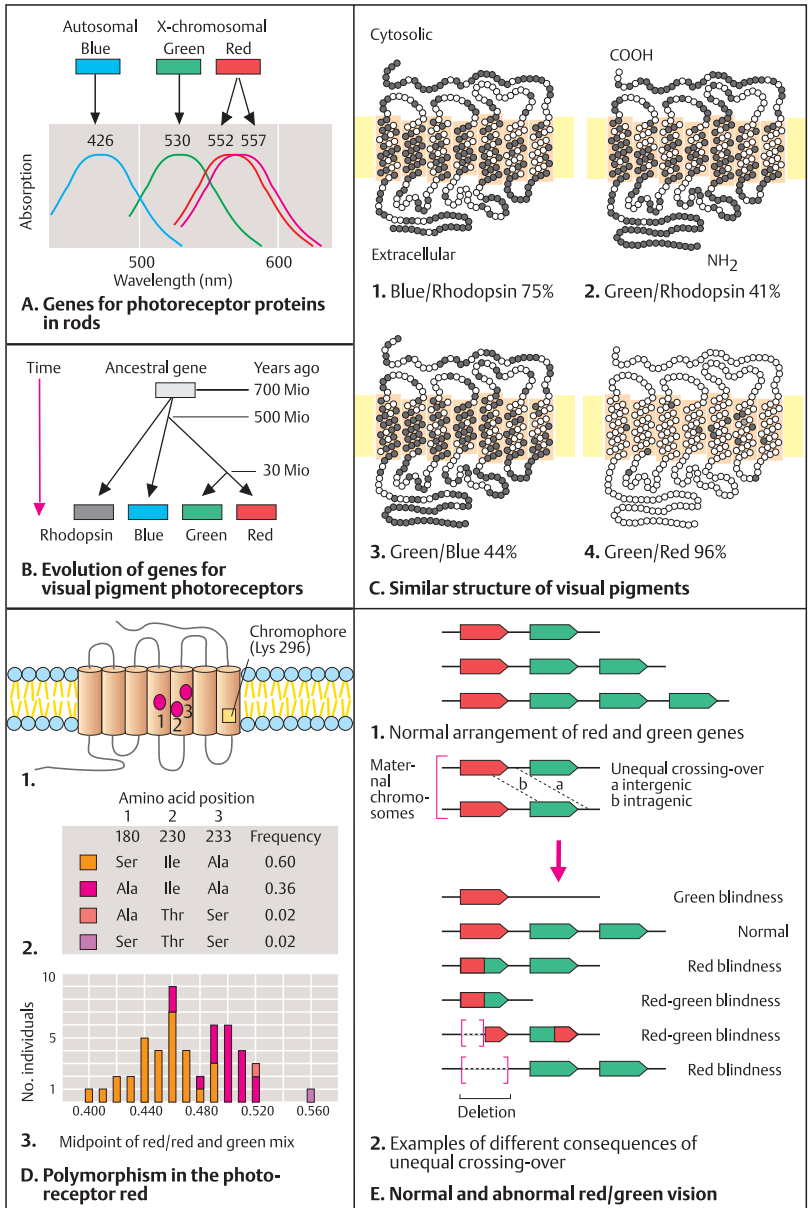
(Ser) (2). Differences in red color perception could be demonstrated by the color-mixing test procedure of Raleigh (3).

E. Normal and defective red-green vision

One gene for red and one to three genes for green lie close together on the long arm of the X chromosome in humans (1). Since the sequences of these genes are very similar, unequal crossing-over is not infrequent (2). Inter-genic crossing-over leads to loss (green blindness) or duplication; intragenic crossing-over leads to a hybrid gene (red blindness). Green blindness results from loss of a gene for the green receptor; red blindness, from a defective or absent red receptor. With red-green blindness, neither a normal red nor a normal green receptor is present. About 1% of all men are red-green blind and about 2% are green blind. About 8% show weakness in differentiating red from green.

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Hearing and Deafness

Acoustic signals are essential for an animal's ability to respond appropriately to its environment. Hearing is orchestrated by a large ensemble of proteins acting in concert. Specialized sensory cells in the cochlea of the inner ear process the incoming sound waves, converting them into cellular information that is relayed to the brain via the acoustic nerve. A missing or defective protein involved in the hearing process results in hearing loss. Hearing loss is common in humans. One out of 1000 newborns lacks the ability to hear or has severely impaired hearing. Two categories of genetic hearing loss can be distinguished: nonsyndromic and syndromic. In the former category the genetic defect is limited to the ear; in the latter the ear is one of several organ systems affected.

The types of genes implicated when defective as the cause of nonsyndromic hearing loss include those encoding proteins involved in cytoskeletal structure, transcription factors, ion channels (potassium channel), and intercellular gap channels composed of junction connexins.

A. The main components of the ear

The auditory system consists of the outer ear, the middle ear, and the inner ear. Sound waves are funneled through the outer ear (auricle) and transmitted through the external ear canal to the tympanic membrane, which they cause to vibrate. These vibrations are transmitted through the tympanic cavity of the middle ear by a chain of three movable bones, the malleus, the incus, and the stapes. Three major cavities form the inner ear: the vestibule, the cochlea, and the semicircular canals. The cochlea is the site where auditory signals are processed. The cochlea contains a membranous labyrinth filled with a fluid, the endolymph. The vestibular apparatus includes three semicircular canals oriented at 90° degree angles to each other. They respond to rotatory and linear acceleration. Signals received here are transmitted via the vestibular nerve, which fuses with the cochlear nerve to form the acoustic nerve. The latter transmits the information to the brain.

B. The cochlea

The cochlea contains the cochlear duct, which forms the organ of Corti. The organ of Corti converts sound waves in the endolymph of the

cochlea into intracellular signals. These are transmitted to auditory regions of the brain. The organ of Corti contains two types of sensory cells: one row of inner hair cells and three rows of outer hair cells. The inner hair cells are pure receptor cells. Vibrations induced by sound lead to slight deflections of the stereocilia and open potassium channels at the tips of the stereocilia. The influx of potassium ions at the tips of the cilia of the hair cells (see C) causes a change in membrane potential that results in a nerve impulse, which is transmitted as an auditory signal to the auditory cortex of the brain. Potassium ions are recycled to the supporting cells and the spiral ligament into the endolymph of the scala media. The tectorial membrane amplifies the sound waves as a resonator. (Figure adapted and redrawn from Willems, 2000.)

C. The outer hair cell

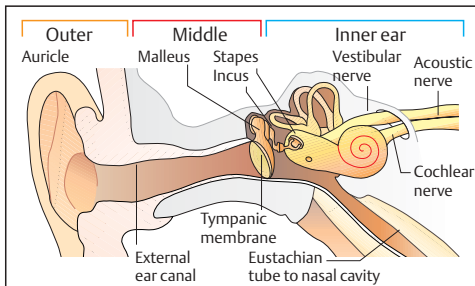
The outer hair cells combine sensory function with the ability to elongate and contract when acoustically stimulated. The apical pole of a hair cell carries an array of about 100 cylindrical stereocilia in a V-shaped arrangement. Each stereocilium contains an actin molecule, which enables it to elongate or to contract. The tips of the stereocilia are connected by tip links. The potassium channels are formed by the KCNQ4 protein (yellow) and by connexins (red). Important for the structural integrity and dynamics of the hair cells is a cytoskeleton involving actin, myosin 7A, myosin 15, and the protein diaphanous. (Figure adapted and redrawn from Willems, 2000.)

D. Chromosomal locations of human deafness genes

Almost every human chromosome harbors at least one gene involved in nonsyndromic monogenic hearing loss. The diagrammatic presentation shown here is limited to nonsyndromic hearing loss.

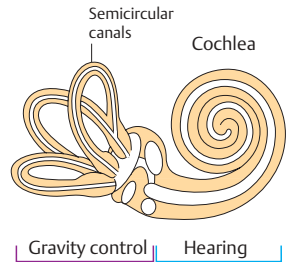
References

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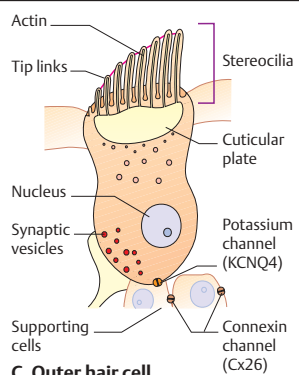
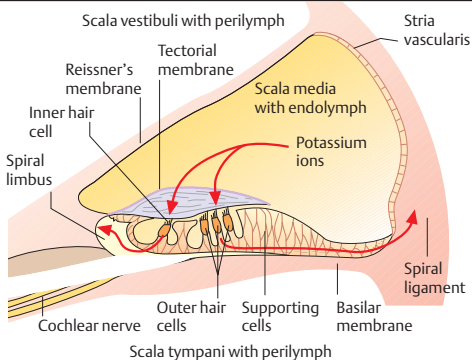


1. Cross section of the human ear

A. The main components of the ear

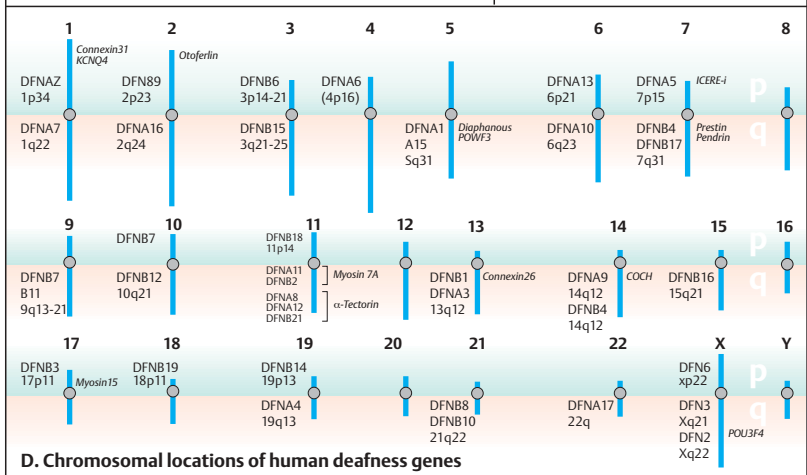


2. Inner ear



B. The cochlea

C. Outer hair cell



Odorant Receptor Gene Family

Vertebrates can differentiate thousands of individual odors. Although their ability to distinguish differences in color is based on only three classes of photoreceptors, their sense of smell is regulated by a large multigene family of receptors that are highly specific for individual odors. In fish, about 100 and in mammals about 1000 genes code for specific olfactory receptors. These genes are expressed exclusively in the olfactory epithelium of the nasal mucous membrane.

A. Olfactory nerve cells in the nasal mucous membrane

The peripheral olfactory neuroepithelium of the nasal mucous membrane consists of three cell types: olfactory sensory neurons, whose axons lead to the olfactory bulb, supporting cells, and basal cells, which serve as stem cells for the formation of olfactory neurons during the individual's entire lifespan. Each olfactory neuron is bipolar, with olfactory cilia in the lumen of the nasal mucous membrane and a projection to the olfactory bulb, the first relay station of the olfactory system on the way to the brain.

B. Odor-specific transmembrane receptors and GTP-binding protein ($G_s[\text{olf}]$)

Each receptor in the cilia of the olfactory neurons binds specifically to one odorant. Binding of the receptor activates adenylate cyclase via a specific GTP-binding protein (stimulatory G protein of the olfactory system, $G_s[\text{olf}]$). This opens a sodium ion channel and initiates a cascade of intracellular signals that result in a nerve signal, which is transmitted to the brain.

C. Olfactory receptor protein

The cloning of a large gene family from the olfactory epithelium of the rat (Buck and Axel, 1991), demonstrated that a receptor protein contains seven transmembrane regions and shows marked structural homology with rhodopsin and β -adrenergic receptors. Unlike rhodopsin, the olfactory receptor proteins contain variable amino acids, especially in the fourth and fifth transmembrane domains. The third intracellular loop between transmembrane domains V and VI is relatively short (17

amino acids), in contrast to other receptor proteins. It is assumed that contact with the various G proteins takes place here (Buck and Axel, 1991).

D. Assignment of olfactory receptor RNA to neurons

A gene for the receptor of a given odorant is expressed in an individual in only a few neurons. Ngai et al. 1993 classified individual olfactory neurons in the olfactory epithelium of the catfish (*Ictalurus punctatus*). Only 0.5–2% of all olfactory neurons recognize a given receptor probe such as probe 202 (1) or 32 (2). Odors are distinguished in the brain according to which neurons are stimulated. The topographical position of each neuron is specific for each odorant.

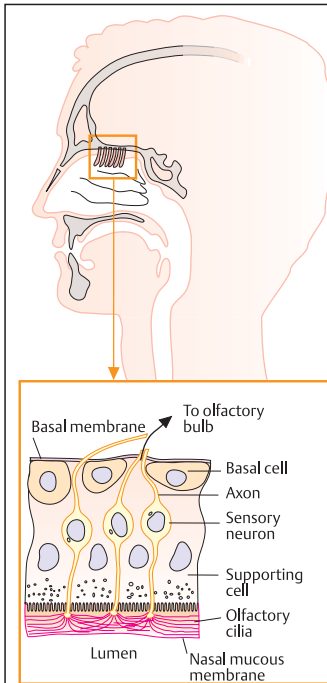
E. Subfamilies within the multigene family

Amino acid sequences derived from partial nucleotide sequences of cDNA clones (F2–F24) (1) investigated by Buck and Axel (1991) were very variable, especially in transmembrane domains III and IV. Within subfamilies, there was homology due to conserved sequences (2). For example, F12 and F13 differ in only 4 of 44 positions (91% identical).

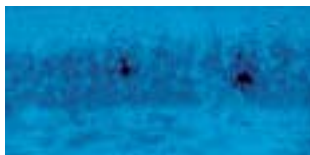
(Figures adapted from Buck & Axel, 1991, and Ngai et al., 1993).

References

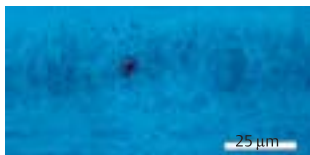
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A. Olfactory nerve cells in the nasal mucous membrane

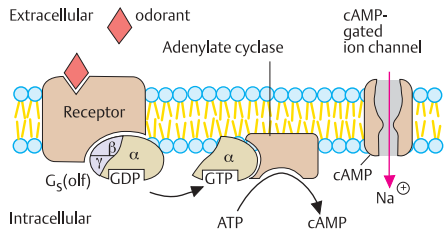


1. Receptor probe 202

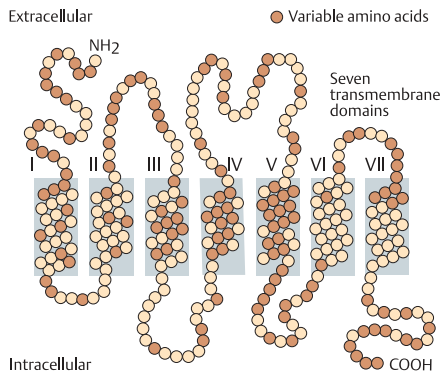


2. Receptor probe 32

D. Correspondence of olfactory receptor RNA to neurons



B. Odor-specific transmembrane receptor and GTP-binding protein [G_s(olf)]



C. Olfactory receptor protein

cDNA clones	Transmembrane domain
F2 RVNE	VV I F I VVSLFLVLPFAL I I MSYV
F3 FLND	LV I YFTLVLLATVPLAGIFYSYF
F5 HLNE	LM I L TEGAVVMVTPFVCIL I SYI
F6 QVVE	LV SFG I AF CV I LGSCG I TLVSYA
F7 HVNE	LV I FVMGG I I LV I PFV L I I VSYV
F8 FPSH	LT MHLVPV I LA A I I SLG I LYSYF
F12 FPSH	LI MN LVPVMLAA I I SFSG I LYSYF
F13 FPSH	LI MN LVPVMLAA I I SFSG I LYSYF
F23 FLND	V I MYFALV L LA VVPLLG I LYSYF
F24 HEIE	MI I LVLA AFNL I SSLLV LVSYL

1. Variable amino acid sequences

F12	FPSH	LI MN LVPVMLAA I I SFSG I LYSYF	KVSSIHS I STVQGKYK
F13	FPSH	LI MN LVPVMLAA I I SFSG I LYSYF	KVSSIRS VSSVKGKYK
F8	FPSH	LT MHLVPV I LA A I I SLG I LYSYF	KVSSIRS MSSVQGKYK
I12	FPSH	LI MN LVPVMLGA I I SFSG I LYSYF	KVSSVRS I SSVQGHKK
F23	FLND	V I MYFALV L LA VVPLLG I LYSYF	KVSSIRA I STVQGKYK
F3	FLND	LV I YFTLVLLATVPLAGIFYSYF	KVSSICA I SSVHGKYK

2. Homology within subfamilies

E. Subfamilies within the multigene family

Mammalian Taste Receptor Gene Family

Aside from the main olfactory system, mammals have evolved two other chemosensory systems, the taste receptor gene family (for bitter taste) and the mammalian pheromone receptor gene family. Five different types of taste can be perceived: salty, sour, bitter, sweet, and umami (the taste of monosodium glutamate, present in Asian food). Salty and sour tastes involve direct effects due to the entry of H^+ and Na^+ ions through specialized membrane channels. In contrast, bitter, sweet, and umami tastes are mediated via a G protein-coupled receptor (GPCR) signaling pathway system. A sweet taste may herald a desirable carbohydrate content, whereas a bitter taste is associated with potentially toxic substances such as alkaloids, cyanides, or other detrimental aromatic compounds.

A. Mammalian chemosensory epithelia

The oral and nasal cavities of mammals contain three distinct chemosensory epithelia: (i) the main olfactory epithelium (MOE) containing sensory cells with odorant receptors in the nose (see previous page), (ii) the taste sensory epithelium of the taste buds of the tongue, soft palate, and epiglottis, and (iii) the vomeronasal organ (VOM, also called Jacobson's organ), a tubular structure in the nasal septum containing sensory cells with pheromone receptors. The main olfactory bulb (MOB) relays signals from the MOE to the olfactory cortex of the brain. The accessory olfactory bulb (AOB) relays signals from the VOM to areas of the amygdala and hypothalamus.

B. Mammalian chemosensory systems

The receptor cells are organized in three corresponding molecular and cellular chemosensory systems. Each neuron of the main olfactory sensory system (1) expresses one of the different olfactory receptor genes and sends axons to specific glomeruli of the main olfactory bulb (mitral cells). The odorant receptor (OR) gene family comprises about 1000 members, each encoding a seven-transmembrane cyclic nucleotide-gated channel with distinct odorant specificity (G-olfactory proteins, G_{olf}). The bitter taste sensory system (2) connects axonal pro-

jections of receptor cells in the taste sensory epithelium of the taste buds to gustatory nuclei of the brain stem. Two families of taste receptors, the TIRs (two genes) and T2Rs (50–80 genes of the gustducin class) have been described. Two families of mammalian putative pheromone receptors (V1Rs and V2Rs) are encoded by 30–50 and over 100 genes, respectively (3).

C. Taste receptor gene family

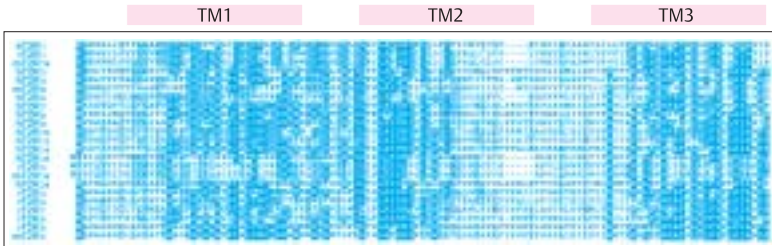
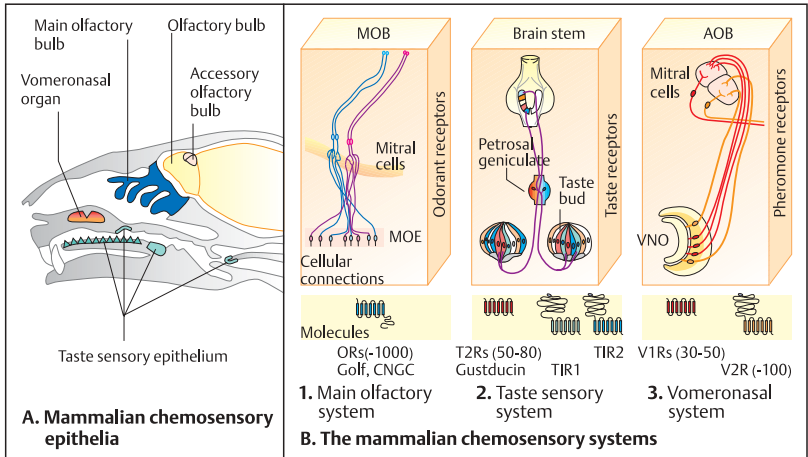
The two novel taste receptor gene families, T1R1 and T1R2, are expressed in distinct subsets of taste receptor cells. The figure shows an alignment of the predicted amino acid sequences of 23 different T2 receptors (T2Rs) of human (h), rat (r), and mouse (m) origin between the first (TM1) and the third (TM3) transmembrane domain. Dark blue indicates identity in at least half of the aligned sequences; light blue represents conserved substitutions; and the remainder are divergent regions. They reflect the ability to bind many structurally different ligands. The T2R genes cluster on a few chromosomes, human chromosomes 5, 7, and 12 and mouse chromosomes 6 and 15.

D. Expression of many taste receptor genes in the same cell

Unlike olfactory system receptor cells, individual taste receptor cells express multiple T2R receptors. Up to ten T2R probes hybridize to only a few cells, shown darkened (1). Double-label fluorescent in-situ hybridization shows that different receptor genes (2, T2R-3 in green and T2R-7 in red, 3) are expressed in the same taste receptor cell. The T2Rs confer high sensitivity for bitter substances at low concentrations but do not distinguish between them. (Figures adapted from Dulac, 2000, and Adler et al., 2000.)

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C. Taste receptor gene family



1. Ten T2R probes



2. T2R-7



3. T2R-3

D. Expression of many taste receptor genes in the same cell

Genes in Embryonic Development

Developmental Mutants in *Drosophila*

The embryonic development of an organism is determined by genes that may be active only during specific phases. Analysis of developmental mutants of embryos of the fruit fly *Drosophila melanogaster* has provided insight into the genetic regulation of developmental processes. Early developmental phases in embryos of very different organisms are regulated by similar genes.

A. The segmental organization of the fruit fly (*Drosophila melanogaster*)

The development of a fruit fly from the fertilized egg cell to the segmented body of the adult organism takes nine days. The larvae hatch after one day and pass through defined stages of embryonic development. The embryo forms a cocoon at five days and, after metamorphosis, emerges as a 2 mm-long adult fly. The head of the adult has three segments (C1–3), the thorax three (T1–3), and the abdomen eight segments (A1–8). A fruit fly has altogether 14 parasegments (P1–14), each corresponding to the last half of one and the first half of the next segment. The segmental organization is discernible in the larva.

B. Embryonic lethal mutations

Embryonic lethal mutations can be identified by appropriate crosses. One-fourth of the progeny of heterozygous flies (A/a) for an embryonic mutant gene (a) are homozygotes (aa) for the mutant allele (1). If a mutation involves maternal genes only (maternal effect), progeny of female homozygotes (bb) are lethally affected (2). Maternal effect genes code for early gene products that determine the polarity of the embryo; see below (C).

C. Examples of developmental mutants

Many developmental mutations are known in *Drosophila melanogaster*. They can be classified into different hierarchical gene classes. The normal larva (wild type) consists of three head seg-

ments, three thorax segments, eight abdominal segments, and the tail end (1). A mutation for anterior maternal effect, *bicoid*, leads to the development of a larva without head or thorax (2). A mutation of a gene called *nanos* affects the posterior end of the early larva. *Gap* genes establish the basic pattern of segmental organization. Mutations of the *gap* genes lead to omissions (gaps) in the segmental construction of the larva. In the *Krippel* mutant (3), all thoracic and the abdominal segments 1–5 are missing; in the *Knirps* mutant (4), abdominal segments 1–6 are absent. The genes for *pair-rule* determine the orientation and developmental fate of the 14 parasegments. Some mutations affect every second segment. With *even-skipped* (5), all even-numbered parasegments are missing. Mutation of the gene *fushi tarazu* leads to fewer than normal segments being formed (*fushi tarazu* is Japanese for too few segments). *Segment polarity* genes determine the polarity of each segment (7). There are more than ten *segment polarity* genes. *Homeotic selector* genes (8) determine the ultimate fate of each segment. With the mutant *antennapedia* (*Ant*), the antenna normally attached immediately under the eye is replaced by a leg (homeotic leg). (Figures adapted from Watson et al., 1992).

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<i>Drosophila melanogaster</i> Adult Segments: C1-3, T1-3, A1-8 Parasegments: 1-3, 4-6, 7-14 Larvae: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 A. Segmental structure of the fruit fly (<i>Drosophila melanogaster</i>)	Head Thorax Abdomen 1. Normal larva 2. <i>bicoid</i> (<i>bcd</i>): no head or thorax 3. <i>Krüppel</i> (<i>kr</i>): all thoracic and abdominal segments 1-5 missing 4. <i>Knirps</i> (<i>kni</i>): abdominal segments 1-6 missing	Gene class Wildtype Anterior maternal effect Gap
1. Embryonic lethal mutants 2. Lethal mutants through maternal effect B. Embryonic lethal mutants	5. <i>even-skipped</i> (<i>eve</i>): all even numbered parasegments missing 6. <i>fushi tarazu</i> (<i>ftz</i>): too few segments; influenced by other genes 7. <i>gooseberry</i> (<i>gb</i>): half of every other segment is missing and replaced by its mirror image 8. Homeotic leg C. Examples of mutants	Pair-rule Segment polarity Homeotic genes

Homeobox Genes

The genes for embryonic development in *Drosophila* are organized in a functional hierarchy. Similar genes (in some regions even identical) occur in mice and in humans.

A. Hierarchy of developmental genes

Four independent systems control the embryonic development of *Drosophila melanogaster*. A gradient of maternally derived protein coded for by the *bicoid* gene (*bcd*) determines development in the anterior region (head); likewise for *nanos* in the posterior region. They activate the *gap* genes. The three most important *gap* genes, *hunchback*, *Knirps*, and *Krüppel*, code for transcription factors of the zinc finger type (p. 218). *Hunchback* is expressed from the anterior part to about the middle of the embryo; *Krüppel* in the region of thoracic segments 4–6 and the first six abdominal segments; *Knirps* further posteriorly (parasegments 10–14). The *gap* genes induce the *pair rule* genes. *Segment polarity* genes determine the correct orientation of the individual segments. After segmentation is complete, *selector* genes determine the further development and the ultimate fate of the segments. They consist of three large gene complexes, the *antennapedia* complex (*ANT-C*), the *bithorax* complex (*BX-C*), and the *ultrabithorax* complex (*UBX*). Mutations of these gene complexes lead to unusual structures.

B. Bithorax mutation

A mutant for the *bithorax* complex (*BX-C*) causes the development of an additional thoracic segment with completely developed wings. (Photograph from Lawrence, 1992, after E. B. Lewis.)

C. Structure of the *antennapedia* gene with homeobox

Antennapedia is a gene of the *antennapedia*-complex (*ANT-C*), which is expressed in parasegments 5 and 6. It contains a segment of highly conserved DNA sequences in exon 8, the homeobox. Its sequence is identical in a wide variety of organisms, from *Drosophila* to mammals. It codes for about 60 amino acids (homeodomain). The homeodomain contains four domains of a helical protein with DNA-binding properties and functions as a transcription factor (Gehring et al., 1990).

D. Homeotic genes in *Drosophila* and *Hox* genes in mouse

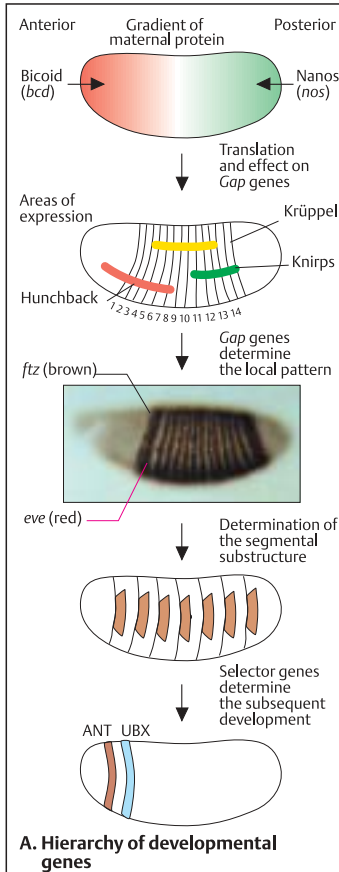
Homeobox genes also occur in vertebrates. A series of genes with the same anterior to posterior orientation and corresponding to the *ANT* and *BX* complexes (*ANT-C* and *BX-C*) are found in the embryonic brain of the mouse (and in man). The temporal expression corresponds to this orientation.

E. Homeobox genes (*HOX* genes)

In humans and in mice there are four groups (clusters) of genes that correspond to the homeo genes of *Drosophila* (*HOX* 1–4 in humans, *Hox* 1–4 in the mouse). Since mutations of these genes in mice lead to characteristic disorders, it is assumed that they are of clinical significance in humans also. The posterior end corresponds to the 5' end and the anterior end to the 3' end of the coding DNA.

References

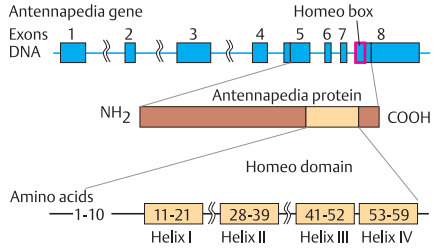
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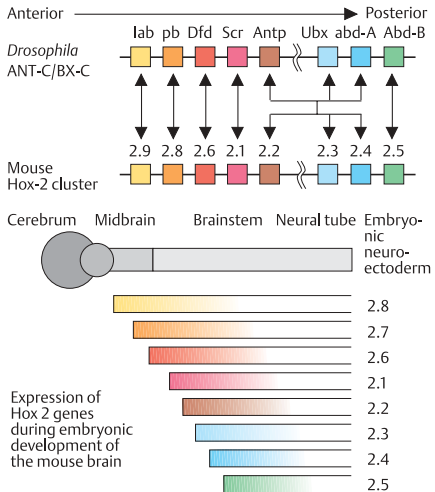
A. Hierarchy of developmental genes



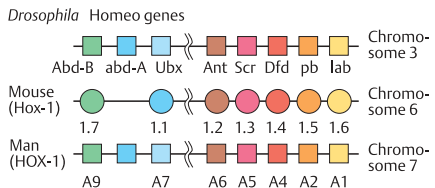
B. Bithorax mutation



C. Structure of the antennapedia gene and Homeobox



D. Homeotic genes in *Drosophila* and Hox genes in the mouse



Further gene clusters in man:

HOX-2 on chromosome 17q21-22 (Mouse chromosome 11)

HOX-3 on chromosome 12q11-21 (Mouse chromosome 15)

HOX-4 on chromosome 2q31-37 (Mouse chromosome 4)

E. Homeobox genes (HOX)

Genetics in a Lucent Vertebrate Embryo: Zebrafish

The zebrafish (*Danio rerio*) is the first vertebrate to be studied by systematic mutation search. The mutations found involve early development, the dorsoventral pattern, formation of the notochord and somites, brain and neural development, the eyes, the ears, the internal organs, blood pigmentation, and other areas. Two of the more than a thousand well-characterized mutants are presented below. Detailed information is available in the zebrafish issue of *Development* (123: 1–481, December 1996).

A. Zebrafish embryos

The zebrafish embryo is optically clear and can easily be studied by dissecting microscopy at maximally 80× magnification. At 29 hours after fertilization (pharyngula period), the main parts of the brain, the forebrain, midbrain, and hindbrain, are clearly visible along with the neural tube, several somites, and the floor plate. At 48 hours (hatching period) pigmentation begins, and the fins, eyes, brain, heart, and other structures become visible. At five days (swimming larva), the outline of a fish begins to show.

B. Induced mutagenesis

Adult fish (1) are used in a crossing scheme involving the exposure of male fish to 3 mM ethylnitrosourea (ENU) in an aqueous solution for three one-hour periods within one week. Mutagenized males are crossed with wild-type females (2). The first generation of this cross (F1) is heterozygous for one mutagenized genome. The next generation (F2) is raised from sibling matings, resulting in the presence of a mutation *m* in 50% of the F2 fish. Random matings involving two heterozygous parents results in 25% homozygous mutant offspring, which are analyzed. A total of 4264 mutants were identified this way by Haffter et al., 1996.

C. Skeletal phenotype of the *fused somites (fss)* mutation

Of the 1163 mutants characterized by their phenotypic effects, two examples are given here and in D. The *fss* mutants show irregular somite boundaries 72 hours after fertilization. At the seven-somite stage the only abnormality is the absence of somite boundaries. At later

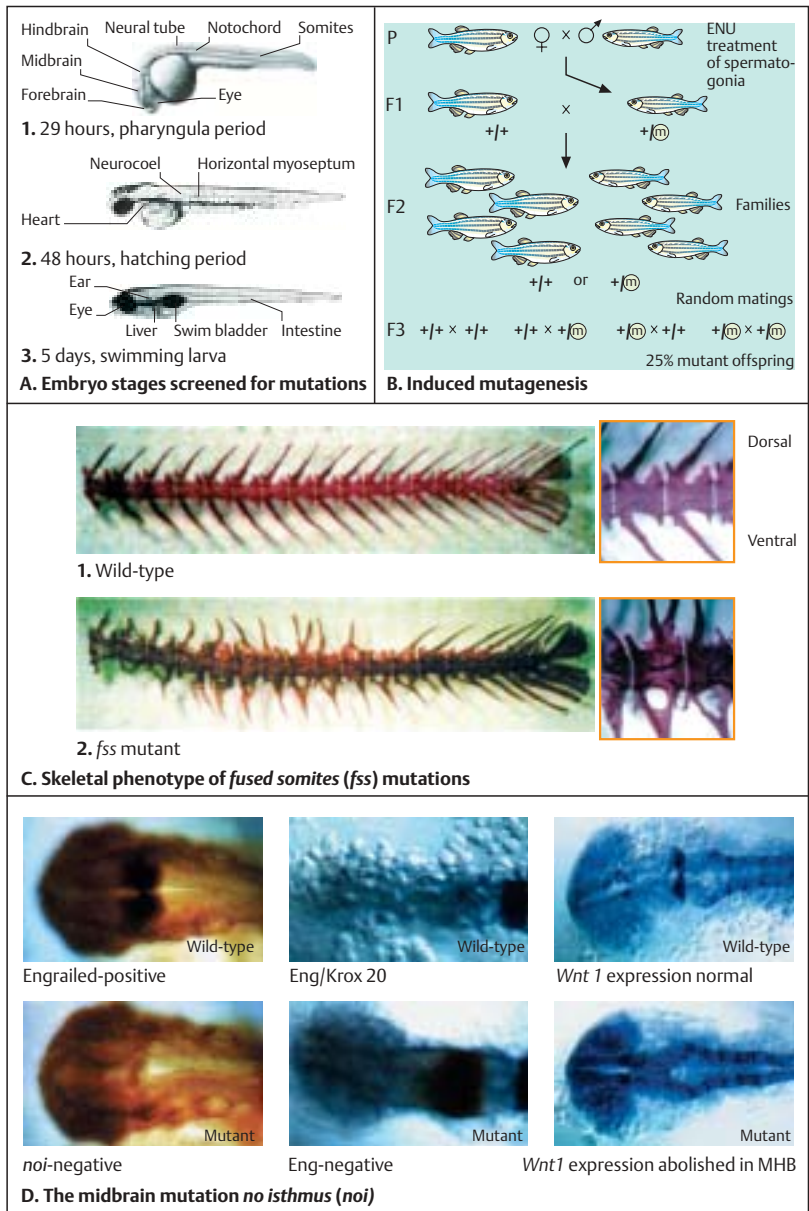
stages malformations of the tail vertebral column are visible (2). In wild-type fish the vertebral centrum has one neural arch dorsally and one hemal arch ventrally. In the *fss* mutant the arches are irregular in shape and, since they grow ectopically, in their relation to each other (2).

D. The midbrain mutation *no isthmus*

This is one example of the more than 60 mutant genes shown to affect the central nervous system and spinal cord (Haffter et al., 1996; Brand et al., 1996). The *no isthmus (noi)* mutation affects the boundary between midbrain and hindbrain. Normal embryos at the 24–48 h pharyngula stage show a conspicuous constriction at this boundary, whereas mutant *noi* embryos do not. In addition, the cerebellum derived from the posterior part is absent. Brand et al. (1996) studied the expression of two genes in this region, *engrailed (eng)* and *wingless (wnt1)*. Normal embryos (wt) at 28 h show strong expression of *eng* between midbrain and hindbrain, while mutant *noi* embryos do not. Eight-somite-stage embryos double stained for *eng* and *krox20* RNA, a marker for rhombomeres 3 and 5 in this region, show *eng* expression and *krox20* expression at the midbrain–hindbrain boundary, but no expression of *eng* in the *noi* mutant. Similarly, *wnt1* expression posterior to the tectum (*tec*, the border of the two brain regions, in the 20-somite-stage embryo is eliminated in the mutant. (Figures adapted from Haffter et al., van Eeden et al., and Brand et al., 1996)

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Developmental Program for Individual Cells in the Nematode *C. elegans*

Development from a zygote into an adult organism requires that the time and place of cell divisions be organized. Whereas cell origin in complex organisms cannot be determined, it is possible to analyze the origin of each individual cell in the nematode *Caenorhabditis elegans*. In 1965, Sydney Brenner introduced the genetic analysis of *C. elegans* as a model to investigate the interaction of genetic, anatomical, and physiological traits in the development of a relatively simple nervous system.

A. *Caenorhabditis elegans*

Caenorhabditis elegans is a small (about 1 mm long), transparent worm with a life cycle of about 3 days. The basic structure is a bilaterally symmetric elongated body of nerves, muscles, skin, and intestine. It exists as one of two sexes: hermaphrodite or male. Hermaphrodites produce eggs and sperm and can reproduce by self-fertilization. The adult hermaphrodite worm has 959 somatic cell nuclei; the adult male worm has 1031. In addition, there are 1000 to 2000 germ cells.

The complete nucleotide sequence of the *C. elegans* genome was reported in 1998. The 97 Mb DNA contains 19099 predicted genes, 12000 without known function. Coding sequences make up about 27% of the DNA, introns account for 26%. About 32% of the coding sequences are similar to human and about 70% of known human proteins have homologies in *C. elegans*. The largest group of genes are transmembrane receptors (790), in particular chemoreceptors; zinc finger transcription factors (480); and proteins with protein-kinase domains. (Diagram from Wood, 1988, after Sulston and Horvitz, 1977.)

B. Embryonic origin of individual cells

The developmental pathway of each individual cell can be traced. The various tissues arise from six founder cells. Of the 959 adult cells, 302 are nerve cells. Except for cells of the intestine and germ line, differentiated tissues stem from different founder cells. Cells with similar function are not necessarily related, while cells with different functions may be of the same origin.

Genetically established rules determine the fate of the two daughter cells at each cell division.

C. Mutations in the developmental control genes

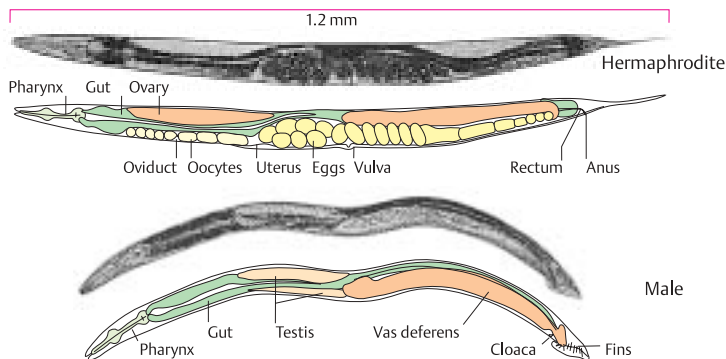
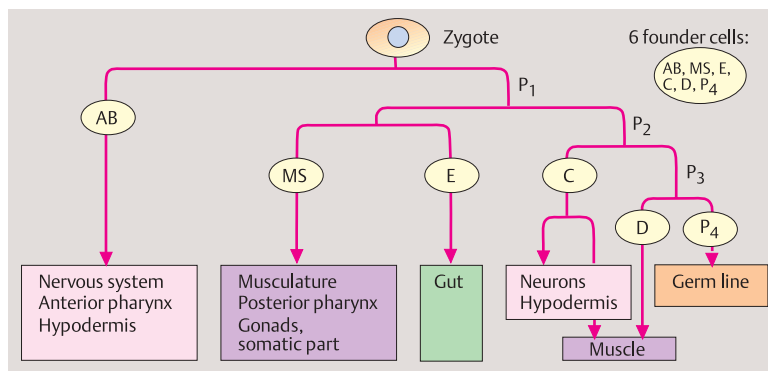
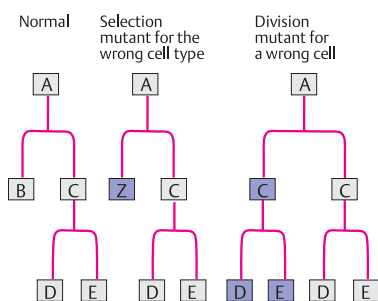
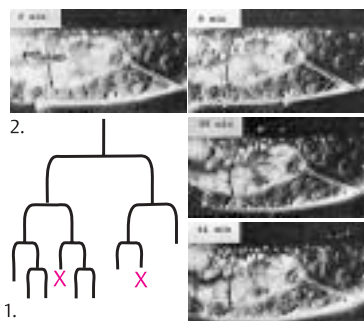
Many developmental control genes have been identified from analysis of ethyl methanesulfonate-induced point mutations. Some mutants determine an incorrect cell type (e.g., Z instead of B); others divide too early or too late (division mutants) resulting, for example, in C twice, instead of B and C.

D. Programmed cell death

Programmed cell death (apoptosis) is a normal part of vertebrate and invertebrate development. During embryonic development of *C. elegans*, one in eight somatic cells regularly dies at a defined time and branching point (1). The photographs (2) show the death of a cell (cell p11.aap) over a time span of about 40 minutes (from Wood, 1988, after Sulston and Horvitz, 1977). Mutations that interfere with programmed cell death can lead to severe disturbances. The human *BCL-2* gene corresponds to the *ced-9* gene in *C. elegans*, a gene that prevents apoptosis. The human *BCL-2* gene codes for an inner mitochondrial membrane protein that acts by inhibiting cell loss by apoptosis in pro-B lymphocytes. Disruption of this gene on human chromosome 18 causes follicular lymphoma, a B-cell tumor.

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**A. *Caenorhabditis elegans*****B. Embryonic origin of individual cells****C. Mutations in developmental control genes****D. Programmed cell death (X)**

Developmental Genes in a Plant Embryo

In plants, as in animals, the basic structural plan is genetically determined. In the plant *Arabidopsis thaliana* (common wall cress), nine genes with numerous alleles determine the organization of the plant embryo along an apical-basal longitudinal axis. They determine a radial pattern and the form (Mayer et al., 1991). This example from plant genetics is included here because *Arabidopsis* is an important object of genetic studies. Its small 130 Mb genome is the first completely sequenced plant (E. Pennisi, Science **290**:32–34, 2000).

The four organs of the flower—sepals, petals, stamens, and carpels—develop under the influence of four classes of genes. Class A genes encode sepals; class A and class B genes together make petals; class B and C induce stamens; and class C genes are required for carpels. The class A and C genes mutually inhibit one another.

Homeotic genes specify the identity of certain tissues in both insects and the organs of the flower of the plant *Arabidopsis*. They are transcribed in a precise spatial pattern and form complex genetic interactions. The overall result is a determined state in which cells maintain a distinct fate irrespective of the environment. Mutations then lead to expression of genes at wrong sites, e.g. legs where antennae should be or petals instead of stamens.

A. Normal development and structure

The basic structural plan can be understood as an axial and a radial pattern superimposed on each other. An octant stage, a globular stage, and a so-called heart stage can be differentiated before the seedling is formed. The regions A, C, and B of the octant stage correspond to the regions A, C, and B of the heart stage. Region A forms the cotyledon and the meristem; C forms, the hypocotyl region; and B forms the root. The seedling consists of a set of identifiable structures including vessels (v), external epidermis (e), short meristem (s), cotyledons (c), and hypocotyl (h), ground tissue (g), and root primordium (r) at the bottom (labeled v instead of r). In the heart stage, the essential organization of the plant is predetermined.

B. Deletions in the apical-basal pattern

The mutations can be induced by 0.3% ethyl methanesulfonate. Using complementation analysis, Mayer et al. (1991) determined mutations in three areas of the plant. These mutations affect the apical-basal pattern, the radial pattern, and the shape. Apical-basal deletions involve one of several genes, each leading to a characteristic phenotype: apical deletion (*Gurke*), central deletion (*Fackel*), basal deletion (*monopteros*), and terminal deletion (*Gnom*).

C. Wild-type

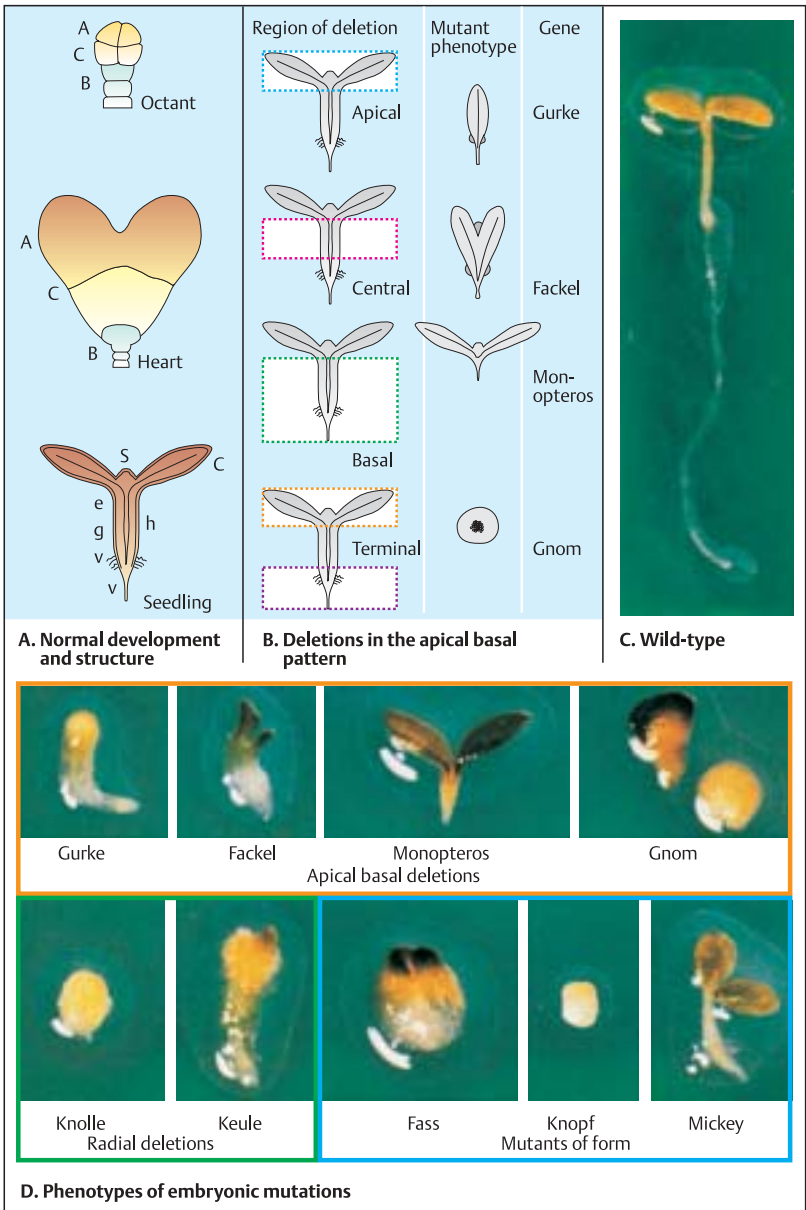
The normal structure of embryonic *Arabidopsis* results from two basic processes: formation of patterns (apical-basal and radial orientation) and morphogenesis through different cell forms and regional differences in cell division.

D. Phenotypes of embryonic mutants

The four mutant phenotypes in the apical-basal pattern are *Gurke* (9 alleles), *Fackel* (5 alleles), *monopteros* (11 alleles), and *Gnom* (15 alleles) (see B). Deletions in the radial pattern lead to phenotypes *Keule* (9 alleles) and *Knolle* (2 alleles). Mutants of shape are *Fass* (12 alleles), *Knopf* (6 alleles), and *Mickey* (8 alleles). (Photographs from Mayer et al., 1991.) The *monopteros* gene (*ml*) is apparently very important for apical-basal development. However, it also has an indirect effect on the spatial arrangement of the apical structures. It is not necessary for root development (Berleth and Jürgens, 1993).

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Immune System

Components of the Immune System

The function of the immune system is to recognize and to eliminate invading foreign organisms. The immune response must be fast, specific, and aimed exclusively against foreign molecules or cells harboring an infective agent, but never against its own normal cells. The effector molecules must be prepared for contact with a variety of organisms and molecular structures. With these requirements, an extremely efficient, genetically complex immune system has evolved. Its development and function are regulated by genes of extraordinary diversity.

A. Lymphatic organs

The immune system consists of peripheral blood lymphocytes and of the lymphatic organs (lymphoid tissue). The primary lymphoid tissues are the thymus and bone marrow. Secondary lymphoid tissues are the lymph nodes in various regions of the body, especially the nasopharynx, axillae, groins, and intestines. The spleen is considered a secondary lymphatic organ.

B. Lymphocytes and the immune response

Lymphocytes carry the immune response. This was shown by the following experiment. The normal immune response of a mouse to an administered antigen (foreign organism, foreign molecular structure) was destroyed by a high dose of X-irradiation, to which the immune system is especially sensitive. The immune response was then restored by lymphocytes of a genetically identical (from an inbred strain) unirradiated mouse, but not by any other cells. Other cells are ineffective since only lymphocytes produce an immune response.

C. T cells and B cells

Lymphocytes exist as one of two functionally and morphologically different types, T lymphocytes and B lymphocytes. T lymphocytes undergo differentiation in the thymus during embryonic and fetal development, thus the designation T lymphocyte or T cell. B lymphocytes

differentiate in the bone marrow in mammals and in the bursa of Fabricius in birds (thus the designation B cells). A series of further differentiating steps take place in the lymph nodes (T cells) and in the spleen (B cells).

D. Cellular and humoral immune response

During the first phase of the immune response induced by an antigen (e.g., a bacterium, virus, fungus, or foreign protein), there is rapid proliferation of B cells (humoral immune response). The B cells mature to plasma cells, which form free antibodies (immunoglobulins) directed at the antigen. The antibodies bind specifically to the antigens. The humoral immune response is rapid, but it does not reach foreign organisms that have invaded body cells. They are the target of the cellular immune response.

E. Antibody molecules (basic structure)

The basic structural motif of an antibody molecule (immunoglobulin) is a Y-shaped protein composed of two heavy chains (H chains) and two light chains (L chains). They are held together at defined sites by disulfide bonds. L chains and H chains contain regions with variable and constant sequences of amino acids.

F. Antigen-antibody binding

A foreign molecule, the antigen, is recognized and firmly bound to a specific region of the antibody molecule, the antigen-binding site. Here, the amino acid sequence of the antibody molecule differs from one molecule to the next in three hypervariable regions. The result is the ability to bind a wide spectrum of different antigenic molecules. The three-dimensional structure of this region is known precisely, and important details of the binding process are understood. (Figures adapted from Alberts et al., 1994).

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